Model for “B20_spinless_Co_spd_Si_sp_all_3_fixed”

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General Condition

- Basis type: `lg`
- SAMB selection:
  - Type: `[Q, G]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A]
  - Spin (s): `[0, 1]`
- Atomic selection:
  - Type: `[Q, G, M, T]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A, E, T]
  - Spin (s): `[0, 1]`
- Site-cluster selection:
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A, E, T]
- Bond-cluster selection:
  - Type: `[Q, G, M, T]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A, E, T]
- Max. neighbor: `3`
- Search cell range: `(-10, 10), (-10, 10), (-10, 10)`
- Toroidal priority: `false`

Q0やQu の中にも、完全にはRとLで一致しないものもありますこの簡単な例をひとつ教えてください

CoSiの Co-Co 1st NN hopping の Q0です。

Left:	**21**	**z21**	**\mathbb{Q}_{0}^{(c)}(A,a)**	**1.67480658E-02**
Right:	**21**	**z21**	**\mathbb{Q}_{0}^{(c)}(A,a)**	**-5.77888528E-03**

Group and Unit Cell

- Group: SG No. 198 T^4 $P2_13$ [cubic]
- Associated point group: PG No. 198 T 23 [cubic]
- Unit cell:
  $a =$ `4.43860`, $b =$ `4.43860`, $c =$ `4.43860`, $\alpha =$ `90.0`, $\beta =$ `90.0`, $\gamma =$ `90.0`
- Lattice vectors (conventional cell):
  $\boldsymbol{a}_1 =$ `[ 4.43860, 0.00000, 0.00000]`
  $\boldsymbol{a}_2 =$ `[ 0.00000, 4.43860, 0.00000]`
  $\boldsymbol{a}_3 =$ `[ 0.00000, 0.00000, 4.43860]`

Symmetry Operation

Table 1: Symmetry operation

	#	SO	#	SO	#	SO	#	SO	#	SO
	1	$\{1\|0\}$	2	$\{2_{001}\|\frac{1}{2}0\frac{1}{2}\}$	3	$\{2_{010}\|0\frac{1}{2}\frac{1}{2}\}$	4	$\{2_{100}\|\frac{1}{2}\frac{1}{2}0\}$	5	$\{3^{+}_{111}\|0\}$
	6	$\{3^{+}_{-11-1}\|\frac{1}{2}\frac{1}{2}0\}$	7	$\{3^{+}_{1-1-1}\|\frac{1}{2}0\frac{1}{2}\}$	8	$\{3^{+}_{-1-11}\|0\frac{1}{2}\frac{1}{2}\}$	9	$\{3^{-}_{111}\|0\}$	10	$\{3^{-}_{1-1-1}\|0\frac{1}{2}\frac{1}{2}\}$
	11	$\{3^{-}_{-1-11}\|\frac{1}{2}\frac{1}{2}0\}$	12	$\{3^{-}_{-11-1}\|\frac{1}{2}0\frac{1}{2}\}$						

Harmonics

Table 2: Harmonics

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
1	$\mathbb{G}_0(A)$	A	0	G, M	-	-	1
2	$\mathbb{Q}_0(A)$	A	0	Q, T	-	-	1
3	$\mathbb{G}_3(A)$	A	3	G, M	-	-	$\sqrt{15}xyz$
4	$\mathbb{Q}_3(A)$	A	3	Q, T	-	-	$\sqrt{15}xyz$
5	$\mathbb{G}_4(A)$	A	4	G, M	-	-	$\frac{\sqrt{21}\left(x^4-3x^2y^2-3x^2z^2+y^4-3y^2z^2+z^4\right)}{6}$
6	$\mathbb{Q}_4(A)$	A	4	Q, T	-	-	$\frac{\sqrt{21}\left(x^4-3x^2y^2-3x^2z^2+y^4-3y^2z^2+z^4\right)}{6}$
7	$\mathbb{Q}_6(A,1)$	A	6	Q, T	1	-	$\frac{\sqrt{2}\left(2x^6-15x^4y^2-15x^4z^2-15x^2y^4+180x^2y^2z^2-15x^2z^4+2y^6-15y^4z^2-15y^2z^4+2z^6\right)}{8}$
8	$\mathbb{Q}_6(A,2)$	A	6	Q, T	2	-	$-\frac{\sqrt{2310}(x-y)(x+y)(x-z)(x+z)(y-z)(y+z)}{8}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
9	$\mathbb{G}_{2,1}(E)$	E	2	G, M	-	1	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$
10	$\mathbb{G}_{2,2}(E)$					2	$\frac{\sqrt{3}(x-y)(x+y)}{2}$
11	$\mathbb{Q}_{2,1}(E)$	E	2	Q, T	-	1	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$
12	$\mathbb{Q}_{2,2}(E)$					2	$\frac{\sqrt{3}(x-y)(x+y)}{2}$
13	$\mathbb{Q}_{4,1}(E)$	E	4	Q, T	-	1	$-\frac{\sqrt{15}\left(x^4-12x^2y^2+6x^2z^2+y^4+6y^2z^2-2z^4\right)}{12}$
14	$\mathbb{Q}_{4,2}(E)$					2	$\frac{\sqrt{5}(x-y)(x+y)\left(x^2+y^2-6z^2\right)}{4}$
15	$\mathbb{G}_{1,1}(T)$	T	1	G, M	-	1	x
16	$\mathbb{G}_{1,2}(T)$					2	y
17	$\mathbb{G}_{1,3}(T)$					3	z
18	$\mathbb{Q}_{1,1}(T)$	T	1	Q, T	-	1	x
19	$\mathbb{Q}_{1,2}(T)$					2	y
20	$\mathbb{Q}_{1,3}(T)$					3	z
21	$\mathbb{G}_{2,1}(T)$	T	2	G, M	-	1	$\sqrt{3}yz$
22	$\mathbb{G}_{2,2}(T)$					2	$\sqrt{3}xz$
23	$\mathbb{G}_{2,3}(T)$					3	$\sqrt{3}xy$
24	$\mathbb{Q}_{2,1}(T)$	T	2	Q, T	-	1	$\sqrt{3}yz$
25	$\mathbb{Q}_{2,2}(T)$					2	$\sqrt{3}xz$
26	$\mathbb{Q}_{2,3}(T)$					3	$\sqrt{3}xy$
27	$\mathbb{G}_{3,1}(T,1)$	T	3	G, M	1	1	$\frac{x\left(2x^2-3y^2-3z^2\right)}{2}$
28	$\mathbb{G}_{3,2}(T,1)$					2	$-\frac{y\left(3x^2-2y^2+3z^2\right)}{2}$
29	$\mathbb{G}_{3,3}(T,1)$					3	$-\frac{z\left(3x^2+3y^2-2z^2\right)}{2}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
30	$\mathbb{G}_{3,1}(T,2)$	T	3	G, M	2	1	$\frac{\sqrt{15}x(y-z)(y+z)}{2}$
31	$\mathbb{G}_{3,2}(T,2)$					2	$-\frac{\sqrt{15}y(x-z)(x+z)}{2}$
32	$\mathbb{G}_{3,3}(T,2)$					3	$\frac{\sqrt{15}z(x-y)(x+y)}{2}$
33	$\mathbb{Q}_{3,1}(T,1)$	T	3	Q, T	1	1	$\frac{x\left(2x^2-3y^2-3z^2\right)}{2}$
34	$\mathbb{Q}_{3,2}(T,1)$					2	$-\frac{y\left(3x^2-2y^2+3z^2\right)}{2}$
35	$\mathbb{Q}_{3,3}(T,1)$					3	$-\frac{z\left(3x^2+3y^2-2z^2\right)}{2}$
36	$\mathbb{Q}_{3,1}(T,2)$	T	3	Q, T	2	1	$\frac{\sqrt{15}x(y-z)(y+z)}{2}$
37	$\mathbb{Q}_{3,2}(T,2)$					2	$-\frac{\sqrt{15}y(x-z)(x+z)}{2}$
38	$\mathbb{Q}_{3,3}(T,2)$					3	$\frac{\sqrt{15}z(x-y)(x+y)}{2}$
39	$\mathbb{Q}_{4,1}(T,1)$	T	4	Q, T	1	1	$\frac{\sqrt{35}yz(y-z)(y+z)}{2}$
40	$\mathbb{Q}_{4,2}(T,1)$					2	$-\frac{\sqrt{35}xz(x-z)(x+z)}{2}$
41	$\mathbb{Q}_{4,3}(T,1)$					3	$\frac{\sqrt{35}xy(x-y)(x+y)}{2}$
42	$\mathbb{Q}_{4,1}(T,2)$	T	4	Q, T	2	1	$\frac{\sqrt{5}yz\left(6x^2-y^2-z^2\right)}{2}$
43	$\mathbb{Q}_{4,2}(T,2)$					2	$-\frac{\sqrt{5}xz\left(x^2-6y^2+z^2\right)}{2}$
44	$\mathbb{Q}_{4,3}(T,2)$					3	$-\frac{\sqrt{5}xy\left(x^2+y^2-6z^2\right)}{2}$

Basis in full matrix

Table 3: dimension = 52

	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)
	0	$\|s\rangle$ @Co(1)	1	$\|p_x\rangle$ @Co(1)	2	$\|p_y\rangle$ @Co(1)	3	$\|p_z\rangle$ @Co(1)	4	$\|d_v\rangle$ @Co(1)
	5	$\|d_{xy}\rangle$ @Co(1)	6	$\|d_{xz}\rangle$ @Co(1)	7	$\|d_{yz}\rangle$ @Co(1)	8	$\|d_u\rangle$ @Co(1)	9	$\|s\rangle$ @Co(2)
	10	$\|p_x\rangle$ @Co(2)	11	$\|p_y\rangle$ @Co(2)	12	$\|p_z\rangle$ @Co(2)	13	$\|d_v\rangle$ @Co(2)	14	$\|d_{xy}\rangle$ @Co(2)
	15	$\|d_{xz}\rangle$ @Co(2)	16	$\|d_{yz}\rangle$ @Co(2)	17	$\|d_u\rangle$ @Co(2)	18	$\|s\rangle$ @Co(3)	19	$\|p_x\rangle$ @Co(3)
	20	$\|p_y\rangle$ @Co(3)	21	$\|p_z\rangle$ @Co(3)	22	$\|d_v\rangle$ @Co(3)	23	$\|d_{xy}\rangle$ @Co(3)	24	$\|d_{xz}\rangle$ @Co(3)
	25	$\|d_{yz}\rangle$ @Co(3)	26	$\|d_u\rangle$ @Co(3)	27	$\|s\rangle$ @Co(4)	28	$\|p_x\rangle$ @Co(4)	29	$\|p_y\rangle$ @Co(4)
	30	$\|p_z\rangle$ @Co(4)	31	$\|d_v\rangle$ @Co(4)	32	$\|d_{xy}\rangle$ @Co(4)	33	$\|d_{xz}\rangle$ @Co(4)	34	$\|d_{yz}\rangle$ @Co(4)
	35	$\|d_u\rangle$ @Co(4)	36	$\|s\rangle$ @Si(1)	37	$\|p_x\rangle$ @Si(1)	38	$\|p_y\rangle$ @Si(1)	39	$\|p_z\rangle$ @Si(1)
	40	$\|s\rangle$ @Si(2)	41	$\|p_x\rangle$ @Si(2)	42	$\|p_y\rangle$ @Si(2)	43	$\|p_z\rangle$ @Si(2)	44	$\|s\rangle$ @Si(3)
	45	$\|p_x\rangle$ @Si(3)	46	$\|p_y\rangle$ @Si(3)	47	$\|p_z\rangle$ @Si(3)	48	$\|s\rangle$ @Si(4)	49	$\|p_x\rangle$ @Si(4)
	50	$\|p_y\rangle$ @Si(4)	51	$\|p_z\rangle$ @Si(4)						

Table 4: Atomic basis (orbital part only)

orbital	definition
$\|s\rangle$	1
$\|p_x\rangle$	x
$\|p_y\rangle$	y
$\|p_z\rangle$	z

continued ...

Table 4

orbital	definition
$\|d_v\rangle$	$\frac{\sqrt{3}(x^2-y^2)}{2}$
$\|d_{xy}\rangle$	$\sqrt{3}xy$
$\|d_{xz}\rangle$	$\sqrt{3}xz$
$\|d_{yz}\rangle$	$\sqrt{3}yz$
$\|d_u\rangle$	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$

SAMB: 164 (all 394)

- `Co` : 'Co' site-cluster
  * bra: $\langle s|$
  * ket: $|s\rangle$
  * wyckoff: `4a`

  z1 $\quad \mathbb{Q}_0^{(c)}(A) = \mathbb{Q}_0^{(a)}(A)\mathbb{Q}_0^{(s)}(A)$

- `Co` : 'Co' site-cluster
  * bra: $\langle s|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: `4a`

  z2 $\quad \mathbb{Q}_0^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(s)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(s)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(s)}(T)}{3}$

- `Co` : 'Co' site-cluster
  * bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: `4a`

  z4 $\quad \mathbb{Q}_0^{(c)}(A) = \mathbb{Q}_0^{(a)}(A)\mathbb{Q}_0^{(s)}(A)$

z5 $\mathbb{Q}_{3}^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(s)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(s)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(s)}(T)}{3}$

- Co : 'Co' site-cluster
  * bra: $\langle s|$
  * ket: $|d_v\rangle$, $|d_{xy}\rangle$, $|d_{xz}\rangle$, $|d_{yz}\rangle$, $|d_u\rangle$
  * wyckoff: 4a

z3 $\mathbb{Q}_{3}^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(s)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(s)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(s)}(T)}{3}$

- Co : 'Co' site-cluster
  * bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
  * ket: $|d_v\rangle$, $|d_{xy}\rangle$, $|d_{xz}\rangle$, $|d_{yz}\rangle$, $|d_u\rangle$
  * wyckoff: 4a

z6 $\mathbb{Q}_{0}^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(s)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(s)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(s)}(T)}{3}$

z7 $\mathbb{Q}_{3}^{(c)}(A) = \mathbb{Q}_{3}^{(a)}(A)\mathbb{Q}_{0}^{(s)}(A)$

z8 $\mathbb{Q}_{4}^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{3,1}^{(a)}(T,1)\mathbb{Q}_{1,1}^{(s)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{3,2}^{(a)}(T,1)\mathbb{Q}_{1,2}^{(s)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{3,3}^{(a)}(T,1)\mathbb{Q}_{1,3}^{(s)}(T)}{3}$

z9 $\mathbb{G}_{3}^{(c)}(A,a) = -\frac{\sqrt{3}\mathbb{Q}_{3,1}^{(a)}(T,2)\mathbb{Q}_{1,1}^{(s)}(T)}{3} - \frac{\sqrt{3}\mathbb{Q}_{3,2}^{(a)}(T,2)\mathbb{Q}_{1,2}^{(s)}(T)}{3} - \frac{\sqrt{3}\mathbb{Q}_{3,3}^{(a)}(T,2)\mathbb{Q}_{1,3}^{(s)}(T)}{3}$

z10 $\mathbb{G}_{3}^{(c)}(A,b) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(s)}(T)}{3} + \frac{\sqrt{3}\mathbb{G}_{2,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(s)}(T)}{3} + \frac{\sqrt{3}\mathbb{G}_{2,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(s)}(T)}{3}$

- Co : 'Co' site-cluster
  * bra: $\langle d_v|$, $\langle d_{xy}|$, $\langle d_{xz}|$, $\langle d_{yz}|$, $\langle d_u|$
  * ket: $|d_v\rangle$, $|d_{xy}\rangle$, $|d_{xz}\rangle$, $|d_{yz}\rangle$, $|d_u\rangle$
  * wyckoff: 4a

z11 $\mathbb{Q}_{0}^{(c)}(A) = \mathbb{Q}_{0}^{(a)}(A)\mathbb{Q}_{0}^{(s)}(A)$

z12 $\mathbb{Q}_{3}^{(c)}(A,a) = \frac{\sqrt{3}\mathbb{Q}_{4,1}^{(a)}(T,2)\mathbb{Q}_{1,1}^{(s)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{4,2}^{(a)}(T,2)\mathbb{Q}_{1,2}^{(s)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{4,3}^{(a)}(T,2)\mathbb{Q}_{1,3}^{(s)}(T)}{3}$

z13 $\mathbb{Q}_{3}^{(c)}(A,b) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(s)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(s)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(s)}(T)}{3}$

z14 $\mathbb{Q}_{4}^{(c)}(A) = \mathbb{Q}_{4}^{(a)}(A)\mathbb{Q}_{0}^{(s)}(A)$

z15 $\mathbb{G}_{4}^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{4,1}^{(a)}(T,1)\mathbb{Q}_{1,1}^{(s)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{4,2}^{(a)}(T,1)\mathbb{Q}_{1,2}^{(s)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{4,3}^{(a)}(T,1)\mathbb{Q}_{1,3}^{(s)}(T)}{3}$

- `Si` : 'Si' site-cluster
  * bra: $\langle s|$
  * ket: $|s\rangle$
  * wyckoff: `4a`

  z16 $\mathbb{Q}_{0}^{(c)}(A) = \mathbb{Q}_{0}^{(a)}(A)\mathbb{Q}_{0}^{(s)}(A)$

- `Si` : 'Si' site-cluster
  * bra: $\langle s|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: `4a`

  z17 $\mathbb{Q}_{0}^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(s)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(s)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(s)}(T)}{3}$

- `Si` : 'Si' site-cluster
  * bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: `4a`

  z18 $\mathbb{Q}_{0}^{(c)}(A) = \mathbb{Q}_{0}^{(a)}(A)\mathbb{Q}_{0}^{(s)}(A)$

  z19 $\mathbb{Q}_{3}^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(s)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(s)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(s)}(T)}{3}$

- `Co;Co_001_1` : 'Co'-'Co' bond-cluster
  * bra: $\langle s|$
  * ket: $|s\rangle$
  * wyckoff: `12a@12b`

  z20 $\mathbb{Q}_{0}^{(c)}(A) = \mathbb{Q}_{0}^{(a)}(A)\mathbb{Q}_{0}^{(b)}(A)$

* common SAMBs

`(Co;Co_001_1, Co;Co_002_1, Co;Co_003_1), (z20, z101, z182)`

- `Co;Co_001_1` : 'Co'-'Co' bond-cluster
  * bra: $\langle s|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: `12a@12b`

z21 $\mathbb{Q}_{0}^{(c)}(A,a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

z22 $\mathbb{Q}_{0}^{(c)}(A,b) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T,a)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T,a)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T,a)}{3}$

z23 $\mathbb{Q}_{0}^{(c)}(A,c) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T,b)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T,b)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T,b)}{3}$

z24 $\mathbb{Q}_{3}^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)}{3}$

z25 $\mathbb{Q}_{4}^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{3,1}^{(b)}(T,1)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{3,2}^{(b)}(T,1)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{3,3}^{(b)}(T,1)}{3}$

z26 $\mathbb{G}_{0}^{(c)}(A) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T)\mathbb{M}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T)\mathbb{M}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T)\mathbb{M}_{1,3}^{(b)}(T)}{3}$

* common SAMBs

`(Co;Co_001_1, Co;Co_002_1, Co;Co_003_1), (z21, z102, z183), (z22, z103, z184), (z23, z104, z185), (z24, z105, z186), (z25, z106, z187), (z26, z107, z188)`

- `Co;Co_001_1` : 'Co'-'Co' bond-cluster
  * bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: `12a@12b`

z37 $\mathbb{Q}_{0}^{(c)}(A,a) = \mathbb{Q}_{0}^{(a)}(A)\mathbb{Q}_{0}^{(b)}(A)$

z38 $\mathbb{Q}_{0}^{(c)}(A,b) = \frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)}{5}$

z39 $\mathbb{Q}_{0}^{(c)}(A,c) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T)\mathbb{M}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T)\mathbb{M}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T)\mathbb{M}_{1,3}^{(b)}(T)}{3}$

z40 $\mathbb{Q}_{3}^{(c)}(A,a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

z41 $\mathbb{Q}_{3}^{(c)}(A,b) = -\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{3,1}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{3,2}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{3,3}^{(b)}(T,1)}{3}$

z42 $\mathbb{Q}_{4}^{(c)}(A) = \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T)}{15} + \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T)}{15} - \frac{\sqrt{30}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)}{15}$

z43 $\mathbb{G}_{0}^{(c)}(A,a) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T,a)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T,a)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T,a)}{3}$

z44 $\mathbb{G}_{0}^{(c)}(A,b) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T,b)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T,b)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T,b)}{3}$

z45 $\mathbb{G}_{3}^{(c)}(A) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{2}$

* common SAMBs

`(Co;Co_001_1, Co;Co_002_1, Co;Co_003_1), (z37, z118, z199), (z38, z119, z200), (z39, z120, z201), (z40, z121, z202), (z41, z122, z203), (z42, z123, z204), (z43, z124, z205), (z44, z125, z206), (z45, z126, z207)`

- `Co;Co_001_1` : 'Co'-'Co' bond-cluster
  * bra: $\langle s|$
  * ket: $|d_v\rangle$, $|d_{xy}\rangle$, $|d_{xz}\rangle$, $|d_{yz}\rangle$, $|d_u\rangle$
  * wyckoff: `12a@12b`

z27 $\mathbb{Q}_{0}^{(c)}(A,a) = \frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)}{5}$

z28 $\mathbb{Q}_{0}^{(c)}(A,b) = \frac{\sqrt{2}\mathbb{T}_{2,1}^{(a)}(E)\mathbb{T}_{2,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(a)}(E)\mathbb{T}_{2,2}^{(b)}(E)}{2}$

z29 $\mathbb{Q}_{3}^{(c)}(A,a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

z30 $\mathbb{Q}_{3}^{(c)}(A,b) = -\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{3,1}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{3,2}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{3,3}^{(b)}(T,1)}{3}$

z31 $\mathbb{Q}_{3}^{(c)}(A,c) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T,a)}{3} + \frac{\sqrt{3}\mathbb{T}_{2,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T,a)}{3} + \frac{\sqrt{3}\mathbb{T}_{2,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T,a)}{3}$

z32 $\mathbb{Q}_{3}^{(c)}(A,d) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T,b)}{3} + \frac{\sqrt{3}\mathbb{T}_{2,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T,b)}{3} + \frac{\sqrt{3}\mathbb{T}_{2,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T,b)}{3}$

z33 $\mathbb{Q}_{4}^{(c)}(A) = \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T)}{15} + \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T)}{15} - \frac{\sqrt{30}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)}{15}$

z34 $\mathbb{G}_{3}^{(c)}(A,a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{2}$

z35 $\mathbb{G}_{3}^{(c)}(A,b) = \frac{\sqrt{2}\mathbb{T}_{2,1}^{(a)}(E)\mathbb{T}_{2,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{T}_{2,2}^{(a)}(E)\mathbb{T}_{2,1}^{(b)}(E)}{2}$

z36 $\mathbb{G}_{3}^{(c)}(A,c) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(a)}(T)\mathbb{M}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{T}_{2,2}^{(a)}(T)\mathbb{M}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{T}_{2,3}^{(a)}(T)\mathbb{M}_{1,3}^{(b)}(T)}{3}$

* common SAMBs

`(Co;Co_001_1, Co;Co_002_1, Co;Co_003_1), (z27, z108, z189), (z28, z109, z190), (z29, z110, z191), (z30, z111, z192), (z31, z112, z193), (z32, z113, z194), (z33, z114, z195), (z34, z115, z196), (z35, z116, z197), (z36, z117, z198)`

- `Co;Co_001_1` : 'Co'-'Co' bond-cluster
  * bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
  * ket: $|d_v\rangle$, $|d_{xy}\rangle$, $|d_{xz}\rangle$, $|d_{yz}\rangle$, $|d_u\rangle$
  * wyckoff: `12a@12b`

z46 $\mathbb{Q}_{0}^{(c)}(A,a) = \frac{\sqrt{3}\mathbb{Q}_{3,1}^{(a)}(T,1)\mathbb{Q}_{3,1}^{(b)}(T,1)}{3} + \frac{\sqrt{3}\mathbb{Q}_{3,2}^{(a)}(T,1)\mathbb{Q}_{3,2}^{(b)}(T,1)}{3} + \frac{\sqrt{3}\mathbb{Q}_{3,3}^{(a)}(T,1)\mathbb{Q}_{3,3}^{(b)}(T,1)}{3}$

z47 $\mathbb{Q}_{0}^{(c)}(A,b) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

z48 $\mathbb{Q}_{0}^{(c)}(A,c) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T,a)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T,a)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T,a)}{3}$

z49 $\mathbb{Q}_{0}^{(c)}(A,d) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T,b)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T,b)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T,b)}{3}$

z50 $\mathbb{Q}_{3}^{(c)}(A,a) = \mathbb{Q}_{3}^{(a)}(A)\mathbb{Q}_{0}^{(b)}(A)$

z51 $\mathbb{Q}_{3}^{(c)}(A,b) = -\frac{\sqrt{3}\mathbb{Q}_{3,1}^{(a)}(T,1)\mathbb{Q}_{2,1}^{(b)}(T)}{3} - \frac{\sqrt{3}\mathbb{Q}_{3,2}^{(a)}(T,1)\mathbb{Q}_{2,2}^{(b)}(T)}{3} - \frac{\sqrt{3}\mathbb{Q}_{3,3}^{(a)}(T,1)\mathbb{Q}_{2,3}^{(b)}(T)}{3}$

z52 $$\mathbb{Q}_{3}^{(c)}(A,c) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)}{3}$$

z53 $$\mathbb{Q}_{3}^{(c)}(A,d) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{G}_{2,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{2}$$

z54 $$\mathbb{Q}_{3}^{(c)}(A,e) = \frac{\sqrt{2}\mathbb{M}_{2,1}^{(a)}(E)\mathbb{T}_{2,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(a)}(E)\mathbb{T}_{2,1}^{(b)}(E)}{2}$$

z55 $$\mathbb{Q}_{3}^{(c)}(A,f) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(a)}(T)\mathbb{M}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{M}_{2,2}^{(a)}(T)\mathbb{M}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{M}_{2,3}^{(a)}(T)\mathbb{M}_{1,3}^{(b)}(T)}{3}$$

z56 $$\mathbb{Q}_{3}^{(c)}(A,g) = \mathbb{T}_{3}^{(a)}(A)\mathbb{T}_{0}^{(b)}(A)$$

z57 $$\mathbb{Q}_{3}^{(c)}(A,h) = -\frac{\sqrt{3}\mathbb{T}_{3,1}^{(a)}(T,2)\mathbb{M}_{1,1}^{(b)}(T)}{3} - \frac{\sqrt{3}\mathbb{T}_{3,2}^{(a)}(T,2)\mathbb{M}_{1,2}^{(b)}(T)}{3} - \frac{\sqrt{3}\mathbb{T}_{3,3}^{(a)}(T,2)\mathbb{M}_{1,3}^{(b)}(T)}{3}$$

z58 $$\mathbb{Q}_{4}^{(c)}(A,a) = \frac{\sqrt{3}\mathbb{Q}_{3,1}^{(a)}(T,1)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{3,2}^{(a)}(T,1)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{3,3}^{(a)}(T,1)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$$

z59 $$\mathbb{Q}_{4}^{(c)}(A,b) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{3,1}^{(b)}(T,1)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{3,2}^{(b)}(T,1)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{3,3}^{(b)}(T,1)}{3}$$

z60 $$\mathbb{Q}_{4}^{(c)}(A,c) = \frac{\sqrt{3}\mathbb{T}_{3,1}^{(a)}(T,1)\mathbb{T}_{1,1}^{(b)}(T,a)}{3} + \frac{\sqrt{3}\mathbb{T}_{3,2}^{(a)}(T,1)\mathbb{T}_{1,2}^{(b)}(T,a)}{3} + \frac{\sqrt{3}\mathbb{T}_{3,3}^{(a)}(T,1)\mathbb{T}_{1,3}^{(b)}(T,a)}{3}$$

z61 $$\mathbb{Q}_{4}^{(c)}(A,d) = \frac{\sqrt{3}\mathbb{T}_{3,1}^{(a)}(T,1)\mathbb{T}_{1,1}^{(b)}(T,b)}{3} + \frac{\sqrt{3}\mathbb{T}_{3,2}^{(a)}(T,1)\mathbb{T}_{1,2}^{(b)}(T,b)}{3} + \frac{\sqrt{3}\mathbb{T}_{3,3}^{(a)}(T,1)\mathbb{T}_{1,3}^{(b)}(T,b)}{3}$$

z62 $$\mathbb{Q}_{6}^{(c)}(A,2) = -\frac{\sqrt{3}\mathbb{Q}_{3,1}^{(a)}(T,2)\mathbb{Q}_{3,1}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{3,2}^{(a)}(T,2)\mathbb{Q}_{3,2}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{3,3}^{(a)}(T,2)\mathbb{Q}_{3,3}^{(b)}(T,1)}{3}$$

z63 $$\mathbb{G}_{0}^{(c)}(A,a) = \frac{\sqrt{5}\mathbb{G}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{G}_{2,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T)}{5} + \frac{\sqrt{5}\mathbb{G}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{G}_{2,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T)}{5} + \frac{\sqrt{5}\mathbb{G}_{2,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)}{5}$$

z64 $$\mathbb{G}_{0}^{(c)}(A,b) = \frac{\sqrt{2}\mathbb{M}_{2,1}^{(a)}(E)\mathbb{T}_{2,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(a)}(E)\mathbb{T}_{2,2}^{(b)}(E)}{2}$$

z65 $$\mathbb{G}_{0}^{(c)}(A,c) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T)\mathbb{M}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T)\mathbb{M}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T)\mathbb{M}_{1,3}^{(b)}(T)}{3}$$

z66 $\mathbb{G}_3^{(c)}(A,a) = -\frac{\sqrt{3}\mathbb{Q}_{3,1}^{(a)}(T,2)\mathbb{Q}_{1,1}^{(b)}(T)}{3} - \frac{\sqrt{3}\mathbb{Q}_{3,2}^{(a)}(T,2)\mathbb{Q}_{1,2}^{(b)}(T)}{3} - \frac{\sqrt{3}\mathbb{Q}_{3,3}^{(a)}(T,2)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

z67 $\mathbb{G}_3^{(c)}(A,b) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{G}_{2,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{G}_{2,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

z68 $\mathbb{G}_3^{(c)}(A,c) = -\frac{\sqrt{3}\mathbb{G}_{2,1}^{(a)}(T)\mathbb{Q}_{3,1}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{G}_{2,2}^{(a)}(T)\mathbb{Q}_{3,2}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{G}_{2,3}^{(a)}(T)\mathbb{Q}_{3,3}^{(b)}(T,1)}{3}$

z69 $\mathbb{G}_3^{(c)}(A,d) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T,a)}{3} + \frac{\sqrt{3}\mathbb{M}_{2,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T,a)}{3} + \frac{\sqrt{3}\mathbb{M}_{2,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T,a)}{3}$

z70 $\mathbb{G}_3^{(c)}(A,e) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T,b)}{3} + \frac{\sqrt{3}\mathbb{M}_{2,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T,b)}{3} + \frac{\sqrt{3}\mathbb{M}_{2,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T,b)}{3}$

z71 $\mathbb{G}_3^{(c)}(A,f) = -\frac{\sqrt{3}\mathbb{T}_{3,1}^{(a)}(T,2)\mathbb{T}_{1,1}^{(b)}(T,a)}{3} - \frac{\sqrt{3}\mathbb{T}_{3,2}^{(a)}(T,2)\mathbb{T}_{1,2}^{(b)}(T,a)}{3} - \frac{\sqrt{3}\mathbb{T}_{3,3}^{(a)}(T,2)\mathbb{T}_{1,3}^{(b)}(T,a)}{3}$

z72 $\mathbb{G}_3^{(c)}(A,g) = -\frac{\sqrt{3}\mathbb{T}_{3,1}^{(a)}(T,2)\mathbb{T}_{1,1}^{(b)}(T,b)}{3} - \frac{\sqrt{3}\mathbb{T}_{3,2}^{(a)}(T,2)\mathbb{T}_{1,2}^{(b)}(T,b)}{3} - \frac{\sqrt{3}\mathbb{T}_{3,3}^{(a)}(T,2)\mathbb{T}_{1,3}^{(b)}(T,b)}{3}$

z73 $\mathbb{G}_4^{(c)}(A,a) = -\frac{\sqrt{3}\mathbb{Q}_{3,1}^{(a)}(T,2)\mathbb{Q}_{2,1}^{(b)}(T)}{3} - \frac{\sqrt{3}\mathbb{Q}_{3,2}^{(a)}(T,2)\mathbb{Q}_{2,2}^{(b)}(T)}{3} - \frac{\sqrt{3}\mathbb{Q}_{3,3}^{(a)}(T,2)\mathbb{Q}_{2,3}^{(b)}(T)}{3}$

z74 $\mathbb{G}_4^{(c)}(A,b) = \frac{\sqrt{30}\mathbb{G}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{G}_{2,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T)}{15} + \frac{\sqrt{30}\mathbb{G}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{G}_{2,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T)}{15} - \frac{\sqrt{30}\mathbb{G}_{2,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)}{15}$

z75 $\mathbb{G}_4^{(c)}(A,c) = \frac{\sqrt{3}\mathbb{T}_{3,1}^{(a)}(T,1)\mathbb{M}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{T}_{3,2}^{(a)}(T,1)\mathbb{M}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{T}_{3,3}^{(a)}(T,1)\mathbb{M}_{1,3}^{(b)}(T)}{3}$

* common SAMBs

(Co;Co_001_1, Co;Co_002_1, Co;Co_003_1), (z46, z127, z208), (z47, z128, z209), (z48, z129, z210), (z49, z130, z211), (z50, z131, z212), (z51, z132, z213), (z52, z133, z214), (z53, z134, z215), (z54, z135, z216), (z55, z136, z217), (z56, z137, z218), (z57, z138, z219), (z58, z139, z220), (z59, z140, z221), (z60, z141, z222), (z61, z142, z223), (z62, z143, z224), (z63, z144, z225), (z64, z145, z226), (z65, z146, z227), (z66, z147, z228), (z67, z148, z229), (z68, z149, z230), (z69, z150, z231), (z70, z151, z232), (z71, z152, z233), (z72, z153, z234), (z73, z154, z235), (z74, z155, z236), (z75, z156, z237)

- Co;Co_001_1 : 'Co'-'Co' bond-cluster
  * bra: $\langle d_v|$, $\langle d_{xy}|$, $\langle d_{xz}|$, $\langle d_{yz}|$, $\langle d_u|$
  * ket: $|d_v\rangle$, $|d_{xy}\rangle$, $|d_{xz}\rangle$, $|d_{yz}\rangle$, $|d_u\rangle$
  * wyckoff: 12a@12b

z76 $\mathbb{Q}_{0}^{(c)}(A,a) = \mathbb{Q}_{0}^{(a)}(A)\mathbb{Q}_{0}^{(b)}(A)$

z77 $\mathbb{Q}_{0}^{(c)}(A,b) = \frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)}{5}$

z78 $\mathbb{Q}_{0}^{(c)}(A,c) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T)\mathbb{M}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T)\mathbb{M}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T)\mathbb{M}_{1,3}^{(b)}(T)}{3}$

z79 $\mathbb{Q}_{3}^{(c)}(A,a) = \frac{\sqrt{3}\mathbb{Q}_{4,1}^{(a)}(T,2)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{4,2}^{(a)}(T,2)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{4,3}^{(a)}(T,2)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

z80 $\mathbb{Q}_{3}^{(c)}(A,b) = -\frac{\sqrt{3}\mathbb{Q}_{4,1}^{(a)}(T,2)\mathbb{Q}_{3,1}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{4,2}^{(a)}(T,2)\mathbb{Q}_{3,2}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{4,3}^{(a)}(T,2)\mathbb{Q}_{3,3}^{(b)}(T,1)}{3}$

z81 $\mathbb{Q}_{3}^{(c)}(A,c) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

z82 $\mathbb{Q}_{3}^{(c)}(A,d) = -\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{3,1}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{3,2}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{3,3}^{(b)}(T,1)}{3}$

z83 $\mathbb{Q}_{3}^{(c)}(A,e) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(a)}(T,2)\mathbb{T}_{1,1}^{(b)}(T,a)}{3} - \frac{\sqrt{3}\mathbb{M}_{3,2}^{(a)}(T,2)\mathbb{T}_{1,2}^{(b)}(T,a)}{3} - \frac{\sqrt{3}\mathbb{M}_{3,3}^{(a)}(T,2)\mathbb{T}_{1,3}^{(b)}(T,a)}{3}$

z84 $\mathbb{Q}_{3}^{(c)}(A,f) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(a)}(T,2)\mathbb{T}_{1,1}^{(b)}(T,b)}{3} - \frac{\sqrt{3}\mathbb{M}_{3,2}^{(a)}(T,2)\mathbb{T}_{1,2}^{(b)}(T,b)}{3} - \frac{\sqrt{3}\mathbb{M}_{3,3}^{(a)}(T,2)\mathbb{T}_{1,3}^{(b)}(T,b)}{3}$

z85 $\mathbb{Q}_{4}^{(c)}(A,a) = \mathbb{Q}_{4}^{(a)}(A)\mathbb{Q}_{0}^{(b)}(A)$

z86 $\mathbb{Q}_{4}^{(c)}(A,b) = \frac{2\sqrt{11}\mathbb{Q}_{4,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{11} - \frac{\sqrt{11}\mathbb{Q}_{4,1}^{(a)}(T,2)\mathbb{Q}_{2,1}^{(b)}(T)}{11} + \frac{2\sqrt{11}\mathbb{Q}_{4,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{11} - \frac{\sqrt{11}\mathbb{Q}_{4,2}^{(a)}(T,2)\mathbb{Q}_{2,2}^{(b)}(T)}{11} - \frac{\sqrt{11}\mathbb{Q}_{4,3}^{(a)}(T,2)\mathbb{Q}_{2,3}^{(b)}(T)}{11}$

z87 $\mathbb{Q}_{4}^{(c)}(A,c) = \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T)}{15} + \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T)}{15} - \frac{\sqrt{30}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)}{15}$

z88 $\mathbb{Q}_{4}^{(c)}(A,d) = \frac{\sqrt{3}\mathbb{M}_{3,1}^{(a)}(T,1)\mathbb{M}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{M}_{3,2}^{(a)}(T,1)\mathbb{M}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{M}_{3,3}^{(a)}(T,1)\mathbb{M}_{1,3}^{(b)}(T)}{3}$

z89 $\mathbb{Q}_{6}^{(c)}(A,1) = \frac{\sqrt{66}\mathbb{Q}_{4,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{22} + \frac{2\sqrt{66}\mathbb{Q}_{4,1}^{(a)}(T,2)\mathbb{Q}_{2,1}^{(b)}(T)}{33} + \frac{\sqrt{66}\mathbb{Q}_{4,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{22} + \frac{2\sqrt{66}\mathbb{Q}_{4,2}^{(a)}(T,2)\mathbb{Q}_{2,2}^{(b)}(T)}{33} + \frac{2\sqrt{66}\mathbb{Q}_{4,3}^{(a)}(T,2)\mathbb{Q}_{2,3}^{(b)}(T)}{33}$

z90 $\mathbb{Q}_{6}^{(c)}(A,2) = -\frac{\sqrt{210}\mathbb{Q}_{4,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{30} - \frac{2\sqrt{10}\mathbb{Q}_{4,1}^{(a)}(T,1)\mathbb{Q}_{2,1}^{(b)}(T)}{15} + \frac{\sqrt{210}\mathbb{Q}_{4,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{30} - \frac{2\sqrt{10}\mathbb{Q}_{4,2}^{(a)}(T,1)\mathbb{Q}_{2,2}^{(b)}(T)}{15} - \frac{2\sqrt{10}\mathbb{Q}_{4,3}^{(a)}(T,1)\mathbb{Q}_{2,3}^{(b)}(T)}{15}$

z91 $\mathbb{G}_{0}^{(c)}(A,a) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T,a)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T,a)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T,a)}{3}$

z92 $\mathbb{G}_{0}^{(c)}(A,b) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T,b)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T,b)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T,b)}{3}$

z93 $\mathbb{G}_{3}^{(c)}(A,a) = -\frac{2\sqrt{15}\mathbb{Q}_{4,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{15} + \frac{\sqrt{35}\mathbb{Q}_{4,1}^{(a)}(T,1)\mathbb{Q}_{2,1}^{(b)}(T)}{15} + \frac{2\sqrt{15}\mathbb{Q}_{4,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{15} + \frac{\sqrt{35}\mathbb{Q}_{4,2}^{(a)}(T,1)\mathbb{Q}_{2,2}^{(b)}(T)}{15} + \frac{\sqrt{35}\mathbb{Q}_{4,3}^{(a)}(T,1)\mathbb{Q}_{2,3}^{(b)}(T)}{15}$

z94 $\mathbb{G}_{3}^{(c)}(A,b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{2}$

z95 $\mathbb{G}_{3}^{(c)}(A,c) = \mathbb{M}_{3}^{(a)}(A)\mathbb{T}_{0}^{(b)}(A)$

z96 $\mathbb{G}_{3}^{(c)}(A,d) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(a)}(T,2)\mathbb{M}_{1,1}^{(b)}(T)}{3} - \frac{\sqrt{3}\mathbb{M}_{3,2}^{(a)}(T,2)\mathbb{M}_{1,2}^{(b)}(T)}{3} - \frac{\sqrt{3}\mathbb{M}_{3,3}^{(a)}(T,2)\mathbb{M}_{1,3}^{(b)}(T)}{3}$

z97 $\mathbb{G}_{4}^{(c)}(A,a) = \frac{\sqrt{3}\mathbb{Q}_{4,1}^{(a)}(T,1)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{4,2}^{(a)}(T,1)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{4,3}^{(a)}(T,1)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

z98 $\mathbb{G}_{4}^{(c)}(A,b) = -\frac{\sqrt{3}\mathbb{Q}_{4,1}^{(a)}(T,1)\mathbb{Q}_{3,1}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{4,2}^{(a)}(T,1)\mathbb{Q}_{3,2}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{4,3}^{(a)}(T,1)\mathbb{Q}_{3,3}^{(b)}(T,1)}{3}$

z99 $\mathbb{G}_{4}^{(c)}(A,c) = \frac{\sqrt{3}\mathbb{M}_{3,1}^{(a)}(T,1)\mathbb{T}_{1,1}^{(b)}(T,a)}{3} + \frac{\sqrt{3}\mathbb{M}_{3,2}^{(a)}(T,1)\mathbb{T}_{1,2}^{(b)}(T,a)}{3} + \frac{\sqrt{3}\mathbb{M}_{3,3}^{(a)}(T,1)\mathbb{T}_{1,3}^{(b)}(T,a)}{3}$

z100 $\mathbb{G}_{4}^{(c)}(A,d) = \frac{\sqrt{3}\mathbb{M}_{3,1}^{(a)}(T,1)\mathbb{T}_{1,1}^{(b)}(T,b)}{3} + \frac{\sqrt{3}\mathbb{M}_{3,2}^{(a)}(T,1)\mathbb{T}_{1,2}^{(b)}(T,b)}{3} + \frac{\sqrt{3}\mathbb{M}_{3,3}^{(a)}(T,1)\mathbb{T}_{1,3}^{(b)}(T,b)}{3}$

* common SAMBs

(Co;Co_001_1, Co;Co_002_1, Co;Co_003_1), (z76, z157, z238), (z77, z158, z239), (z78, z159, z240), (z79, z160, z241), (z80, z161, z242), (z81, z162, z243), (z82, z163, z244), (z83, z164, z245), (z84, z165, z246), (z85, z166, z247), (z86, z167, z248), (z87, z168, z249), (z88, z169, z250), (z89, z170, z251), (z90, z171, z252), (z91, z172, z253), (z92, z173, z254), (z93, z174, z255), (z94, z175, z256), (z95, z176, z257), (z96, z177, z258), (z97, z178, z259), (z98, z179, z260), (z99, z180, z261), (z100, z181, z262)

- Si;Co_001_1 : 'Co'-'Si' bond-cluster
  * bra: $\langle s|$

* ket: $|s\rangle$
* wyckoff: `4a@4a`

z263 $\mathbb{Q}_0^{(c)}(A) = \mathbb{Q}_0^{(a)}(A)\mathbb{Q}_0^{(b)}(A)$

- `Si;Co_001_1` : 'Co'-'Si' bond-cluster
  * bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
  * ket: $|s\rangle$
  * wyckoff: `4a@4a`

  z265 $\mathbb{Q}_0^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

- `Si;Co_001_1` : 'Co'-'Si' bond-cluster
  * bra: $\langle d_v|$, $\langle d_{xy}|$, $\langle d_{xz}|$, $\langle d_{yz}|$, $\langle d_u|$
  * ket: $|s\rangle$
  * wyckoff: `4a@4a`

  z269 $\mathbb{Q}_3^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

- `Si;Co_001_1` : 'Co'-'Si' bond-cluster
  * bra: $\langle s|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: `4a@4a`

  z264 $\mathbb{Q}_0^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

- `Si;Co_001_1` : 'Co'-'Si' bond-cluster
  * bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: `4a@4a`

  z266 $\mathbb{Q}_0^{(c)}(A) = \mathbb{Q}_0^{(a)}(A)\mathbb{Q}_0^{(b)}(A)$

  z267 $\mathbb{Q}_3^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

  z268 $\mathbb{G}_0^{(c)}(A) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T)}{3}$

- `Si;Co_001_1` : 'Co'-'Si' bond-cluster
  * bra: $\langle d_v|$, $\langle d_{xy}|$, $\langle d_{xz}|$, $\langle d_{yz}|$, $\langle d_u|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: `4a@4a`

  z270 $\mathbb{Q}_0^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

  z271 $\mathbb{Q}_3^{(c)}(A) = \mathbb{Q}_3^{(a)}(A)\mathbb{Q}_0^{(b)}(A)$

  z272 $\mathbb{Q}_4^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{3,1}^{(a)}(T,1)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{3,2}^{(a)}(T,1)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{3,3}^{(a)}(T,1)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

  z273 $\mathbb{G}_3^{(c)}(A,a) = -\frac{\sqrt{3}\mathbb{Q}_{3,1}^{(a)}(T,2)\mathbb{Q}_{1,1}^{(b)}(T)}{3} - \frac{\sqrt{3}\mathbb{Q}_{3,2}^{(a)}(T,2)\mathbb{Q}_{1,2}^{(b)}(T)}{3} - \frac{\sqrt{3}\mathbb{Q}_{3,3}^{(a)}(T,2)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

  z274 $\mathbb{G}_3^{(c)}(A,b) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{G}_{2,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{G}_{2,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

- `Si;Co_002_1` : 'Co'-'Si' bond-cluster
  * bra: $\langle s|$
  * ket: $|s\rangle$
  * wyckoff: `12a@12b`

  z275 $\mathbb{Q}_0^{(c)}(A) = \mathbb{Q}_0^{(a)}(A)\mathbb{Q}_0^{(b)}(A)$

* common SAMBs

`(Si;Co_002_1, Si;Co_003_1), (z275, z311)`

- `Si;Co_002_1` : 'Co'-'Si' bond-cluster
  * bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
  * ket: $|s\rangle$
  * wyckoff: `12a@12b`

  z279 $\mathbb{Q}_0^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

  z280 $\mathbb{Q}_3^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)}{3}$

  z281 $\mathbb{Q}_4^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{3,1}^{(b)}(T,1)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{3,2}^{(b)}(T,1)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{3,3}^{(b)}(T,1)}{3}$

* common SAMBs

(Si;Co_002_1, Si;Co_003_1), (z279, z315), (z280, z316), (z281, z317)

- Si;Co_002_1 : 'Co'-'Si' bond-cluster
  * bra: $\langle d_v|$, $\langle d_{xy}|$, $\langle d_{xz}|$, $\langle d_{yz}|$, $\langle d_u|$
  * ket: $|s\rangle$
  * wyckoff: 12a@12b

z291 $\mathbb{Q}_0^{(c)}(A) = \frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)}{5}$

z292 $\mathbb{Q}_3^{(c)}(A,a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

z293 $\mathbb{Q}_3^{(c)}(A,b) = -\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{3,1}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{3,2}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{3,3}^{(b)}(T,1)}{3}$

z294 $\mathbb{Q}_4^{(c)}(A) = \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T)}{15} + \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T)}{15} - \frac{\sqrt{30}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)}{15}$

z295 $\mathbb{G}_3^{(c)}(A) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{2}$

* common SAMBs

(Si;Co_002_1, Si;Co_003_1), (z291, z327), (z292, z328), (z293, z329), (z294, z330), (z295, z331)

- Si;Co_002_1 : 'Co'-'Si' bond-cluster
  * bra: $\langle s|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: 12a@12b

z276 $\mathbb{Q}_0^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

z277 $\mathbb{Q}_3^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)}{3}$

z278 $\mathbb{Q}_4^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{3,1}^{(b)}(T,1)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{3,2}^{(b)}(T,1)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{3,3}^{(b)}(T,1)}{3}$

* common SAMBs

(Si;Co_002_1, Si;Co_003_1), (z276, z312), (z277, z313), (z278, z314)

- Si;Co_002_1 : 'Co'-'Si' bond-cluster
  * bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: 12a@12b

z282 $\mathbb{Q}_{0}^{(c)}(A,a) = \mathbb{Q}_{0}^{(a)}(A)\mathbb{Q}_{0}^{(b)}(A)$

z283 $\mathbb{Q}_{0}^{(c)}(A,b) = \frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)}{5}$

z284 $\mathbb{Q}_{0}^{(c)}(A,c) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T)\mathbb{M}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T)\mathbb{M}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T)\mathbb{M}_{1,3}^{(b)}(T)}{3}$

z285 $\mathbb{Q}_{3}^{(c)}(A,a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

z286 $\mathbb{Q}_{3}^{(c)}(A,b) = -\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{3,1}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{3,2}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{3,3}^{(b)}(T,1)}{3}$

z287 $\mathbb{Q}_{4}^{(c)}(A) = \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T)}{15} + \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T)}{15} - \frac{\sqrt{30}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)}{15}$

z288 $\mathbb{G}_{0}^{(c)}(A,a) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T,a)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T,a)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T,a)}{3}$

z289 $\mathbb{G}_{0}^{(c)}(A,b) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T,b)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T,b)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T,b)}{3}$

z290 $\mathbb{G}_{3}^{(c)}(A) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{2}$

* common SAMBs

(Si;Co_002_1, Si;Co_003_1), (z282, z318), (z283, z319), (z284, z320), (z285, z321), (z286, z322), (z287, z323), (z288, z324), (z289, z325), (z290, z326)

- Si;Co_002_1 : 'Co'-'Si' bond-cluster
  * bra: $\langle d_v|$, $\langle d_{xy}|$, $\langle d_{xz}|$, $\langle d_{yz}|$, $\langle d_u|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$

* wyckoff: `12a@12b`

z296 $$\mathbb{Q}_{0}^{(c)}(A,a) = \frac{\sqrt{3}\mathbb{Q}_{3,1}^{(a)}(T,1)\mathbb{Q}_{3,1}^{(b)}(T,1)}{3} + \frac{\sqrt{3}\mathbb{Q}_{3,2}^{(a)}(T,1)\mathbb{Q}_{3,2}^{(b)}(T,1)}{3} + \frac{\sqrt{3}\mathbb{Q}_{3,3}^{(a)}(T,1)\mathbb{Q}_{3,3}^{(b)}(T,1)}{3}$$

z297 $$\mathbb{Q}_{0}^{(c)}(A,b) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$$

z298 $$\mathbb{Q}_{3}^{(c)}(A,a) = \mathbb{Q}_{3}^{(a)}(A)\mathbb{Q}_{0}^{(b)}(A)$$

z299 $$\mathbb{Q}_{3}^{(c)}(A,b) = -\frac{\sqrt{3}\mathbb{Q}_{3,1}^{(a)}(T,1)\mathbb{Q}_{2,1}^{(b)}(T)}{3} - \frac{\sqrt{3}\mathbb{Q}_{3,2}^{(a)}(T,1)\mathbb{Q}_{2,2}^{(b)}(T)}{3} - \frac{\sqrt{3}\mathbb{Q}_{3,3}^{(a)}(T,1)\mathbb{Q}_{2,3}^{(b)}(T)}{3}$$

z300 $$\mathbb{Q}_{3}^{(c)}(A,c) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)}{3}$$

z301 $$\mathbb{Q}_{3}^{(c)}(A,d) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{G}_{2,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{2}$$

z302 $$\mathbb{Q}_{4}^{(c)}(A,a) = \frac{\sqrt{3}\mathbb{Q}_{3,1}^{(a)}(T,1)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{3,2}^{(a)}(T,1)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{3,3}^{(a)}(T,1)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$$

z303 $$\mathbb{Q}_{4}^{(c)}(A,b) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{3,1}^{(b)}(T,1)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{3,2}^{(b)}(T,1)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{3,3}^{(b)}(T,1)}{3}$$

z304 $$\mathbb{Q}_{6}^{(c)}(A,2) = -\frac{\sqrt{3}\mathbb{Q}_{3,1}^{(a)}(T,2)\mathbb{Q}_{3,1}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{3,2}^{(a)}(T,2)\mathbb{Q}_{3,2}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{3,3}^{(a)}(T,2)\mathbb{Q}_{3,3}^{(b)}(T,1)}{3}$$

z305 $$\mathbb{G}_{0}^{(c)}(A) = \frac{\sqrt{5}\mathbb{G}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{G}_{2,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T)}{5} + \frac{\sqrt{5}\mathbb{G}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{G}_{2,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T)}{5} + \frac{\sqrt{5}\mathbb{G}_{2,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)}{5}$$

z306 $$\mathbb{G}_{3}^{(c)}(A,a) = -\frac{\sqrt{3}\mathbb{Q}_{3,1}^{(a)}(T,2)\mathbb{Q}_{1,1}^{(b)}(T)}{3} - \frac{\sqrt{3}\mathbb{Q}_{3,2}^{(a)}(T,2)\mathbb{Q}_{1,2}^{(b)}(T)}{3} - \frac{\sqrt{3}\mathbb{Q}_{3,3}^{(a)}(T,2)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$$

z307 $$\mathbb{G}_{3}^{(c)}(A,b) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{G}_{2,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{G}_{2,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$$

z308 $$\mathbb{G}_{3}^{(c)}(A,c) = -\frac{\sqrt{3}\mathbb{G}_{2,1}^{(a)}(T)\mathbb{Q}_{3,1}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{G}_{2,2}^{(a)}(T)\mathbb{Q}_{3,2}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{G}_{2,3}^{(a)}(T)\mathbb{Q}_{3,3}^{(b)}(T,1)}{3}$$

z309 $\mathbb{G}_4^{(c)}(A,a) = -\frac{\sqrt{3}\mathbb{Q}_{3,1}^{(a)}(T,2)\mathbb{Q}_{2,1}^{(b)}(T)}{3} - \frac{\sqrt{3}\mathbb{Q}_{3,2}^{(a)}(T,2)\mathbb{Q}_{2,2}^{(b)}(T)}{3} - \frac{\sqrt{3}\mathbb{Q}_{3,3}^{(a)}(T,2)\mathbb{Q}_{2,3}^{(b)}(T)}{3}$

z310 $\mathbb{G}_4^{(c)}(A,b) = \frac{\sqrt{30}\mathbb{G}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{G}_{2,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T)}{15} + \frac{\sqrt{30}\mathbb{G}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{G}_{2,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T)}{15} - \frac{\sqrt{30}\mathbb{G}_{2,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)}{15}$

* common SAMBs

(Si;Co_002_1, Si;Co_003_1), (z296, z332), (z297, z333), (z298, z334), (z299, z335), (z300, z336), (z301, z337), (z302, z338), (z303, z339), (z304, z340), (z305, z341), (z306, z342), (z307, z343), (z308, z344), (z309, z345), (z310, z346)

- Si;Si_001_1 : 'Si'-'Si' bond-cluster
  * bra: $\langle s|$
  * ket: $|s\rangle$
  * wyckoff: 12a@12b

z347 $\mathbb{Q}_0^{(c)}(A) = \mathbb{Q}_0^{(a)}(A)\mathbb{Q}_0^{(b)}(A)$

* common SAMBs

(Si;Si_001_1, Si;Si_002_1, Si;Si_003_1), (z347, z363, z379)

- Si;Si_001_1 : 'Si'-'Si' bond-cluster
  * bra: $\langle s|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: 12a@12b

z348 $\mathbb{Q}_0^{(c)}(A,a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

z349 $\mathbb{Q}_0^{(c)}(A,b) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T,a)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T,a)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T,a)}{3}$

z350 $\mathbb{Q}_0^{(c)}(A,c) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T,b)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T,b)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T,b)}{3}$

z351 $\mathbb{Q}_3^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)}{3}$

z352 $\mathbb{Q}_4^{(c)}(A) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{3,1}^{(b)}(T,1)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{3,2}^{(b)}(T,1)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{3,3}^{(b)}(T,1)}{3}$

z353 $\mathbb{G}_0^{(c)}(A) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T)\mathbb{M}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T)\mathbb{M}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T)\mathbb{M}_{1,3}^{(b)}(T)}{3}$

* common SAMBs

(Si;Si_001_1, Si;Si_002_1, Si;Si_003_1), (z348, z364, z380), (z349, z365, z381), (z350, z366, z382), (z351, z367, z383), (z352, z368, z384), (z353, z369, z385)

- Si;Si_001_1 : 'Si'-'Si' bond-cluster
  * bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: 12a@12b

z354 $\mathbb{Q}_{0}^{(c)}(A,a) = \mathbb{Q}_{0}^{(a)}(A)\mathbb{Q}_{0}^{(b)}(A)$

z355 $\mathbb{Q}_{0}^{(c)}(A,b) = \frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)}{5}$

z356 $\mathbb{Q}_{0}^{(c)}(A,c) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T)\mathbb{M}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T)\mathbb{M}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T)\mathbb{M}_{1,3}^{(b)}(T)}{3}$

z357 $\mathbb{Q}_{3}^{(c)}(A,a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(b)}(T)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(b)}(T)}{3}$

z358 $\mathbb{Q}_{3}^{(c)}(A,b) = -\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{3,1}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{3,2}^{(b)}(T,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{3,3}^{(b)}(T,1)}{3}$

z359 $\mathbb{Q}_{4}^{(c)}(A) = \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T)}{15} + \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{10} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T)}{15} - \frac{\sqrt{30}\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)}{15}$

z360 $\mathbb{G}_{0}^{(c)}(A,a) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T,a)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T,a)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T,a)}{3}$

z361 $\mathbb{G}_{0}^{(c)}(A,b) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T,b)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T,b)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T,b)}{3}$

z362 $\mathbb{G}_{3}^{(c)}(A) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)}{2}$

* common SAMBs

(Si;Si_001_1, Si;Si_002_1, Si;Si_003_1), (z354, z370, z386), (z355, z371, z387), (z356, z372, z388), (z357, z373, z389), (z358, z374, z390), (z359, z375, z391), (z360, z376, z392), (z361, z377, z393), (z362, z378, z394)

- bra: $\langle s|$
- ket: $|s\rangle$

x1 $\quad \mathbb{Q}_0^{(a)}(A) = \begin{bmatrix} 1 \end{bmatrix}$

- bra: $\langle s|$
- ket: $|p_x\rangle\,,\ |p_y\rangle\,,\ |p_z\rangle$

x2 $\quad \mathbb{Q}_{1,1}^{(a)}(T) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & 0 \end{bmatrix}$

x3 $\quad \mathbb{Q}_{1,2}^{(a)}(T) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$

x4 $\quad \mathbb{Q}_{1,3}^{(a)}(T) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$

x5 $\quad \mathbb{T}_{1,1}^{(a)}(T) = \begin{bmatrix} \frac{\sqrt{2}i}{2} & 0 & 0 \end{bmatrix}$

x6 $\quad \mathbb{T}_{1,2}^{(a)}(T) = \begin{bmatrix} 0 & \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$

x7 $\quad \mathbb{T}_{1,3}^{(a)}(T) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{2} \end{bmatrix}$

- bra: $\langle p_x|\,,\ \langle p_y|\,,\ \langle p_z|$
- ket: $|p_x\rangle\,,\ |p_y\rangle\,,\ |p_z\rangle$

x8 $\quad \mathbb{Q}_0^{(a)}(A) = \begin{bmatrix} \frac{\sqrt{3}}{3} & 0 & 0 \\ 0 & \frac{\sqrt{3}}{3} & 0 \\ 0 & 0 & \frac{\sqrt{3}}{3} \end{bmatrix}$

x9 $\quad \mathbb{Q}_{2,1}^{(a)}(E) = \begin{bmatrix} -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{3} \end{bmatrix}$

x10 $\quad \mathbb{Q}_{2,2}^{(a)}(E) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$

x11 $\mathbb{Q}^{(a)}_{2,1}(T) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$

x12 $\mathbb{Q}^{(a)}_{2,2}(T) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{2} \\ 0 & 0 & 0 \\ \frac{\sqrt{2}}{2} & 0 & 0 \end{bmatrix}$

x13 $\mathbb{Q}^{(a)}_{2,3}(T) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

x14 $\mathbb{M}^{(a)}_{1,1}(T) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{2} \\ 0 & \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$

x15 $\mathbb{M}^{(a)}_{1,2}(T) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{2} & 0 & 0 \end{bmatrix}$

x16 $\mathbb{M}^{(a)}_{1,3}(T) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} & 0 \\ \frac{\sqrt{2}i}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

- bra: $\langle s|$
- ket: $|d_v\rangle$, $|d_{xy}\rangle$, $|d_{xz}\rangle$, $|d_{yz}\rangle$, $|d_u\rangle$

x17 $\mathbb{Q}^{(a)}_{2,1}(E) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$

x18 $\mathbb{Q}^{(a)}_{2,2}(E) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$

x19 $\mathbb{Q}^{(a)}_{2,1}(T) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$

x20 $\mathbb{Q}^{(a)}_{2,2}(T) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{2} & 0 & 0 \end{bmatrix}$

x21 $\mathbb{Q}^{(a)}_{2,3}(T) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & 0 & 0 & 0 \end{bmatrix}$

x22 $\mathbb{T}^{(a)}_{2,1}(E) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{2} \end{bmatrix}$

x23 $\mathbb{T}^{(a)}_{2,2}(E) = \begin{bmatrix} \frac{\sqrt{2}i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$

x24 $\mathbb{T}^{(a)}_{2,1}(T) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$

x25 $\mathbb{T}^{(a)}_{2,2}(T) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{2} & 0 & 0 \end{bmatrix}$

x26 $\mathbb{T}^{(a)}_{2,3}(T) = \begin{bmatrix} 0 & \frac{\sqrt{2}i}{2} & 0 & 0 & 0 \end{bmatrix}$

- bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
- ket: $|d_v\rangle$, $|d_{xy}\rangle$, $|d_{xz}\rangle$, $|d_{yz}\rangle$, $|d_u\rangle$

x27 $\mathbb{Q}^{(a)}_{3}(A) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \end{bmatrix}$

x28 $\mathbb{Q}^{(a)}_{1,1}(T) = \begin{bmatrix} \frac{\sqrt{15}}{10} & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{15}}{10} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}}{10} & 0 & 0 \end{bmatrix}$

x29 $\mathbb{Q}^{(a)}_{1,2}(T) = \begin{bmatrix} 0 & \frac{\sqrt{15}}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & 0 & 0 & \frac{\sqrt{15}}{10} & 0 \end{bmatrix}$

x30 $\mathbb{Q}^{(a)}_{1,3}(T) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{15}}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{10} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{5} \end{bmatrix}$

x31 $\mathbb{Q}^{(a)}_{3,1}(T,1) = \begin{bmatrix} \frac{3\sqrt{10}}{20} & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 \end{bmatrix}$

x32 $\mathbb{Q}^{(a)}_{3,2}(T,1) = \begin{bmatrix} 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 \\ -\frac{3\sqrt{10}}{20} & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 \end{bmatrix}$

x33 $\mathbb{Q}^{(a)}_{3,3}(T,1) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{10} \end{bmatrix}$

x34 $\mathbb{Q}^{(a)}_{3,1}(T,2) = \begin{bmatrix} -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 \end{bmatrix}$

x35 $\mathbb{Q}^{(a)}_{3,2}(T,2) = \begin{bmatrix} 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \end{bmatrix}$

x36 $\mathbb{Q}^{(a)}_{3,3}(T,2) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 \\ \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \end{bmatrix}$

x37 $\mathbb{G}^{(a)}_{2,1}(E) = \begin{bmatrix} 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

x38 $\mathbb{G}^{(a)}_{2,2}(E) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{3} & 0 & 0 & 0 \end{bmatrix}$

x39 $\mathbb{G}^{(a)}_{2,1}(T) = \begin{bmatrix} -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & -\frac{1}{2} \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \end{bmatrix}$

x40 $\mathbb{G}^{(a)}_{2,2}(T) = \begin{bmatrix} 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 \end{bmatrix}$

x41 $\mathbb{G}^{(a)}_{2,3}(T) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{3}}{3} & 0 & 0 & 0 & 0 \end{bmatrix}$

x42 $\mathbb{M}^{(a)}_{2,1}(E) = \begin{bmatrix} 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & \frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

x43 $\mathbb{M}^{(a)}_{2,2}(E) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{3} & 0 & 0 & 0 \end{bmatrix}$

x44 $\mathbb{M}^{(a)}_{2,1}(T) = \begin{bmatrix} \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & \frac{i}{2} \\ 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 \end{bmatrix}$

x45 $\mathbb{M}^{(a)}_{2,2}(T) = \begin{bmatrix} 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 \end{bmatrix}$

x46 $\mathbb{M}^{(a)}_{2,3}(T) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ -\frac{\sqrt{3}i}{3} & 0 & 0 & 0 & 0 \end{bmatrix}$

x47 $\mathbb{T}_{3}^{(a)}(A) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \end{bmatrix}$

x48 $\mathbb{T}_{1,1}^{(a)}(T) = \begin{bmatrix} \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} \\ 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 \end{bmatrix}$

x49 $\mathbb{T}_{1,2}^{(a)}(T) = \begin{bmatrix} 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 \end{bmatrix}$

x50 $\mathbb{T}_{1,3}^{(a)}(T) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} \end{bmatrix}$

x51 $\mathbb{T}_{3,1}^{(a)}(T,1) = \begin{bmatrix} \frac{3\sqrt{10}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} \\ 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 \end{bmatrix}$

x52 $\mathbb{T}_{3,2}^{(a)}(T,1) = \begin{bmatrix} 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ -\frac{3\sqrt{10}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 \end{bmatrix}$

x53 $\mathbb{T}_{3,3}^{(a)}(T,1) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{10} \end{bmatrix}$

x54 $\mathbb{T}_{3,1}^{(a)}(T,2) = \begin{bmatrix} -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 \end{bmatrix}$

x55 $\mathbb{T}^{(a)}_{3,2}(T,2) = \begin{bmatrix} 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$

x56 $\mathbb{T}^{(a)}_{3,3}(T,2) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 \end{bmatrix}$

- bra: $\langle d_v|$, $\langle d_{xy}|$, $\langle d_{xz}|$, $\langle d_{yz}|$, $\langle d_u|$
- ket: $|d_v\rangle$, $|d_{xy}\rangle$, $|d_{xz}\rangle$, $|d_{yz}\rangle$, $|d_u\rangle$

x57 $\mathbb{Q}^{(a)}_{0}(A) = \begin{bmatrix} \frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{5} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}}{5} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{5} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{5} \end{bmatrix}$

x58 $\mathbb{Q}^{(a)}_{4}(A) = \begin{bmatrix} \frac{\sqrt{30}}{10} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{15} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{15} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{15} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{10} \end{bmatrix}$

x59 $\mathbb{Q}^{(a)}_{2,1}(E) = \begin{bmatrix} -\frac{\sqrt{14}}{7} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}}{7} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{7} \end{bmatrix}$

x60 $\mathbb{Q}^{(a)}_{2,2}(E) = \begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{7} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{14} & 0 \\ -\frac{\sqrt{14}}{7} & 0 & 0 & 0 & 0 \end{bmatrix}$

x61 $\mathbb{Q}^{(a)}_{4,1}(E) = \begin{bmatrix} -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{42}}{21} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{21} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} \end{bmatrix}$

x62 $\mathbb{Q}^{(a)}_{4,2}(E) = \begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{14} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{7} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{7} & 0 \\ -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 \end{bmatrix}$

x63 $\mathbb{Q}^{(a)}_{2,1}(T) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{42}}{14} & 0 \\ 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 \\ -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & \frac{\sqrt{14}}{14} \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \end{bmatrix}$

x64 $\mathbb{Q}^{(a)}_{2,2}(T) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 \\ \frac{\sqrt{42}}{14} & 0 & 0 & 0 & \frac{\sqrt{14}}{14} \\ 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \end{bmatrix}$

x65 $\mathbb{Q}^{(a)}_{2,3}(T) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{7} \\ 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 \\ 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & -\frac{\sqrt{14}}{7} & 0 & 0 & 0 \end{bmatrix}$

x66 $\mathbb{Q}^{(a)}_{4,1}(T,1) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & -\frac{\sqrt{6}}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{4} & 0 \end{bmatrix}$

x67 $\mathbb{Q}^{(a)}_{4,2}(T,1) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & \frac{\sqrt{6}}{4} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{4} & 0 & 0 \end{bmatrix}$

x68 $\mathbb{Q}^{(a)}_{4,3}(T,1) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & 0 & 0 & 0 \\ \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

x69 $\mathbb{Q}^{(a)}_{4,1}(T,2) = \begin{bmatrix} 0 & 0 & 0 & \frac{3\sqrt{14}}{28} & 0 \\ 0 & 0 & \frac{\sqrt{14}}{7} & 0 & 0 \\ 0 & \frac{\sqrt{14}}{7} & 0 & 0 & 0 \\ \frac{3\sqrt{14}}{28} & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 \end{bmatrix}$

x70 $\mathbb{Q}_{4,2}^{(a)}(T,2) = \begin{bmatrix} 0 & 0 & -\frac{3\sqrt{14}}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{7} & 0 \\ -\frac{3\sqrt{14}}{28} & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} \\ 0 & \frac{\sqrt{14}}{7} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 \end{bmatrix}$

x71 $\mathbb{Q}_{4,3}^{(a)}(T,2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} \\ 0 & 0 & 0 & \frac{\sqrt{14}}{7} & 0 \\ 0 & 0 & \frac{\sqrt{14}}{7} & 0 & 0 \\ 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 \end{bmatrix}$

x72 $\mathbb{M}_{3}^{(a)}(A) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$

x73 $\mathbb{M}_{1,1}^{(a)}(T) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{10} & 0 \end{bmatrix}$

x74 $\mathbb{M}_{1,2}^{(a)}(T) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 \\ -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & \frac{\sqrt{30}i}{10} \\ 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{10} & 0 & 0 \end{bmatrix}$

x75 $\mathbb{M}_{1,3}^{(a)}(T) = \begin{bmatrix} 0 & -\frac{\sqrt{10}i}{5} & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{5} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

x76 $\mathbb{M}_{3,1}^{(a)}(T,1) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{5} & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{5} & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 \end{bmatrix}$

x77 $\mathbb{M}_{3,2}^{(a)}(T,1) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{5} & 0 \\ \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} \\ 0 & -\frac{\sqrt{10}i}{5} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 \end{bmatrix}$

x78 $\mathbb{M}_{3,3}^{(a)}(T,1) = \begin{bmatrix} 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{5} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{5} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

x79 $\mathbb{M}_{3,1}^{(a)}(T,2) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{6}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{4} & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$

x80 $\mathbb{M}^{(a)}_{3,2}(T,2) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{4} & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \end{bmatrix}$

x81 $\mathbb{M}^{(a)}_{3,3}(T,2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{2} & 0 & 0 & 0 \end{bmatrix}$

Cluster SAMB

- Site cluster

** Wyckoff: `4a`

y1 $\mathbb{Q}^{(s)}_{0}(A) = \left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$

y2 $\mathbb{Q}^{(s)}_{1,1}(T) = \left[\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}\right]$

y3 $\mathbb{Q}^{(s)}_{1,2}(T) = \left[\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right]$

y4 $\mathbb{Q}^{(s)}_{1,3}(T) = \left[\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}\right]$

- Bond cluster

** Wyckoff: `4a@4a`

y5 $\mathbb{Q}^{(s)}_{0}(A) = \left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$

y6 $\mathbb{T}_{0}^{(s)}(A) = \left[\frac{i}{2}, \frac{i}{2}, \frac{i}{2}, \frac{i}{2}\right]$

y7 $\mathbb{Q}_{1,1}^{(s)}(T) = \left[\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}\right]$

y8 $\mathbb{Q}_{1,2}^{(s)}(T) = \left[\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right]$

y9 $\mathbb{Q}_{1,3}^{(s)}(T) = \left[\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}\right]$

y10 $\mathbb{T}_{1,1}^{(s)}(T) = \left[\frac{i}{2}, -\frac{i}{2}, -\frac{i}{2}, \frac{i}{2}\right]$

y11 $\mathbb{T}_{1,2}^{(s)}(T) = \left[\frac{i}{2}, -\frac{i}{2}, \frac{i}{2}, -\frac{i}{2}\right]$

y12 $\mathbb{T}_{1,3}^{(s)}(T) = \left[\frac{i}{2}, \frac{i}{2}, -\frac{i}{2}, -\frac{i}{2}\right]$

** Wyckoff: `12a@12b`

y13 $\mathbb{Q}_{0}^{(s)}(A) = \left[\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}\right]$

y14 $\mathbb{T}_{0}^{(s)}(A) = \left[\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}\right]$

y15 $\mathbb{Q}_{2,1}^{(s)}(E) = \left[-\frac{5\sqrt{2778}}{1389}, -\frac{5\sqrt{2778}}{1389}, -\frac{5\sqrt{2778}}{1389}, -\frac{5\sqrt{2778}}{1389}, -\frac{23\sqrt{2778}}{5556}, -\frac{23\sqrt{2778}}{5556}, -\frac{23\sqrt{2778}}{5556}, -\frac{23\sqrt{2778}}{5556}, \frac{43\sqrt{2778}}{5556}, \frac{43\sqrt{2778}}{5556}, \frac{43\sqrt{2778}}{5556}, \frac{43\sqrt{2778}}{5556}\right]$

y16 $\mathbb{Q}_{2,2}^{(s)}(E) = \left[\frac{11\sqrt{926}}{926}, \frac{11\sqrt{926}}{926}, \frac{11\sqrt{926}}{926}, \frac{11\sqrt{926}}{926}, -\frac{21\sqrt{926}}{1852}, -\frac{21\sqrt{926}}{1852}, -\frac{21\sqrt{926}}{1852}, -\frac{21\sqrt{926}}{1852}, -\frac{\sqrt{926}}{1852}, -\frac{\sqrt{926}}{1852}, -\frac{\sqrt{926}}{1852}, -\frac{\sqrt{926}}{1852}\right]$

y17 $\mathbb{T}_{2,1}^{(s)}(E) = \left[-\frac{5\sqrt{2778}i}{1389}, -\frac{5\sqrt{2778}i}{1389}, -\frac{5\sqrt{2778}i}{1389}, -\frac{5\sqrt{2778}i}{1389}, -\frac{23\sqrt{2778}i}{5556}, -\frac{23\sqrt{2778}i}{5556}, -\frac{23\sqrt{2778}i}{5556}, -\frac{23\sqrt{2778}i}{5556}, \frac{43\sqrt{2778}i}{5556}, \frac{43\sqrt{2778}i}{5556}, \frac{43\sqrt{2778}i}{5556}, \frac{43\sqrt{2778}i}{5556}\right]$

y18 $\mathbb{T}_{2,2}^{(s)}(E) = \left[\frac{11\sqrt{926}i}{926}, \frac{11\sqrt{926}i}{926}, \frac{11\sqrt{926}i}{926}, \frac{11\sqrt{926}i}{926}, -\frac{21\sqrt{926}i}{1852}, -\frac{21\sqrt{926}i}{1852}, -\frac{21\sqrt{926}i}{1852}, -\frac{21\sqrt{926}i}{1852}, -\frac{\sqrt{926}i}{1852}, -\frac{\sqrt{926}i}{1852}, -\frac{\sqrt{926}i}{1852}, -\frac{\sqrt{926}i}{1852}\right]$

y19 $\mathbb{Q}_{1,1}^{(s)}(T) = \left[\frac{5\sqrt{2}}{16}, -\frac{5\sqrt{2}}{16}, -\frac{5\sqrt{2}}{16}, \frac{5\sqrt{2}}{16}, \frac{\sqrt{2}}{8}, \frac{\sqrt{2}}{8}, -\frac{\sqrt{2}}{8}, -\frac{\sqrt{2}}{8}, \frac{\sqrt{6}}{16}, -\frac{\sqrt{6}}{16}, \frac{\sqrt{6}}{16}, -\frac{\sqrt{6}}{16}\right]$

y20 $\mathbb{Q}_{1,2}^{(s)}(T) = \left[\frac{\sqrt{6}}{16}, -\frac{\sqrt{6}}{16}, \frac{\sqrt{6}}{16}, -\frac{\sqrt{6}}{16}, \frac{5\sqrt{2}}{16}, -\frac{5\sqrt{2}}{16}, -\frac{5\sqrt{2}}{16}, \frac{5\sqrt{2}}{16}, \frac{\sqrt{2}}{8}, \frac{\sqrt{2}}{8}, -\frac{\sqrt{2}}{8}, -\frac{\sqrt{2}}{8}\right]$

y21 $\mathbb{Q}_{1,3}^{(s)}(T) = \left[\frac{\sqrt{2}}{8}, \frac{\sqrt{2}}{8}, -\frac{\sqrt{2}}{8}, -\frac{\sqrt{2}}{8}, \frac{\sqrt{6}}{16}, -\frac{\sqrt{6}}{16}, \frac{\sqrt{6}}{16}, -\frac{\sqrt{6}}{16}, \frac{5\sqrt{2}}{16}, -\frac{5\sqrt{2}}{16}, -\frac{5\sqrt{2}}{16}, \frac{5\sqrt{2}}{16}\right]$

y22 $\mathbb{T}_{1,1}^{(s)}(T,a) = \left[\frac{5\sqrt{2}i}{16}, -\frac{5\sqrt{2}i}{16}, -\frac{5\sqrt{2}i}{16}, \frac{5\sqrt{2}i}{16}, \frac{\sqrt{2}i}{8}, \frac{\sqrt{2}i}{8}, -\frac{\sqrt{2}i}{8}, -\frac{\sqrt{2}i}{8}, \frac{\sqrt{6}i}{16}, -\frac{\sqrt{6}i}{16}, \frac{\sqrt{6}i}{16}, -\frac{\sqrt{6}i}{16}\right]$

y23 $\mathbb{T}_{1,2}^{(s)}(T,a) = \left[\frac{\sqrt{6}i}{16}, -\frac{\sqrt{6}i}{16}, \frac{\sqrt{6}i}{16}, -\frac{\sqrt{6}i}{16}, \frac{5\sqrt{2}i}{16}, -\frac{5\sqrt{2}i}{16}, -\frac{5\sqrt{2}i}{16}, \frac{5\sqrt{2}i}{16}, \frac{\sqrt{2}i}{8}, \frac{\sqrt{2}i}{8}, -\frac{\sqrt{2}i}{8}, -\frac{\sqrt{2}i}{8}\right]$

y24 $\mathbb{T}_{1,3}^{(s)}(T,a) = \left[\frac{\sqrt{2}i}{8}, \frac{\sqrt{2}i}{8}, -\frac{\sqrt{2}i}{8}, -\frac{\sqrt{2}i}{8}, \frac{\sqrt{6}i}{16}, -\frac{\sqrt{6}i}{16}, \frac{\sqrt{6}i}{16}, -\frac{\sqrt{6}i}{16}, \frac{5\sqrt{2}i}{16}, -\frac{5\sqrt{2}i}{16}, -\frac{5\sqrt{2}i}{16}, \frac{5\sqrt{2}i}{16}\right]$

y25 $\mathbb{T}_{1,1}^{(s)}(T,b) = \left[\frac{5\sqrt{821}i}{3284}, -\frac{5\sqrt{821}i}{3284}, -\frac{5\sqrt{821}i}{3284}, \frac{5\sqrt{821}i}{3284}, -\frac{11\sqrt{821}i}{821}, -\frac{11\sqrt{821}i}{821}, \frac{11\sqrt{821}i}{821}, \frac{11\sqrt{821}i}{821}, \frac{21\sqrt{2463}i}{3284}, -\frac{21\sqrt{2463}i}{3284}, \frac{21\sqrt{2463}i}{3284}, -\frac{21\sqrt{2463}i}{3284}\right]$

y26 $\mathbb{T}_{1,2}^{(s)}(T,b) = \left[\frac{21\sqrt{2463}i}{3284}, -\frac{21\sqrt{2463}i}{3284}, \frac{21\sqrt{2463}i}{3284}, -\frac{21\sqrt{2463}i}{3284}, \frac{5\sqrt{821}i}{3284}, -\frac{5\sqrt{821}i}{3284}, -\frac{5\sqrt{821}i}{3284}, \frac{5\sqrt{821}i}{3284}, -\frac{11\sqrt{821}i}{821}, -\frac{11\sqrt{821}i}{821}, \frac{11\sqrt{821}i}{821}, \frac{11\sqrt{821}i}{821}\right]$

y27 $\mathbb{T}_{1,3}^{(s)}(T,b) = \left[-\frac{11\sqrt{821}i}{821}, -\frac{11\sqrt{821}i}{821}, \frac{11\sqrt{821}i}{821}, \frac{11\sqrt{821}i}{821}, \frac{21\sqrt{2463}i}{3284}, -\frac{21\sqrt{2463}i}{3284}, \frac{21\sqrt{2463}i}{3284}, -\frac{21\sqrt{2463}i}{3284}, \frac{5\sqrt{821}i}{3284}, -\frac{5\sqrt{821}i}{3284}, -\frac{5\sqrt{821}i}{3284}, \frac{5\sqrt{821}i}{3284}\right]$

y28 $\mathbb{M}_{1,1}^{(s)}(T) = \left[-\frac{43\sqrt{4926}i}{13136}, \frac{43\sqrt{4926}i}{13136}, \frac{43\sqrt{4926}i}{13136}, -\frac{43\sqrt{4926}i}{13136}, \frac{25\sqrt{4926}i}{6568}, \frac{25\sqrt{4926}i}{6568}, -\frac{25\sqrt{4926}i}{6568}, -\frac{25\sqrt{4926}i}{6568}, \frac{115\sqrt{1642}i}{13136}, -\frac{115\sqrt{1642}i}{13136}, \frac{115\sqrt{1642}i}{13136}, -\frac{115\sqrt{1642}i}{13136}\right]$

y29 $\mathbb{M}_{1,2}^{(s)}(T) = \left[\frac{115\sqrt{1642}i}{13136}, -\frac{115\sqrt{1642}i}{13136}, \frac{115\sqrt{1642}i}{13136}, -\frac{115\sqrt{1642}i}{13136}, -\frac{43\sqrt{4926}i}{13136}, \frac{43\sqrt{4926}i}{13136}, \frac{43\sqrt{4926}i}{13136}, -\frac{43\sqrt{4926}i}{13136}, \frac{25\sqrt{4926}i}{6568}, \frac{25\sqrt{4926}i}{6568}, -\frac{25\sqrt{4926}i}{6568}, -\frac{25\sqrt{4926}i}{6568}\right]$

y30 $\mathbb{M}_{1,3}^{(s)}(T) = \left[\frac{25\sqrt{4926}i}{6568}, \frac{25\sqrt{4926}i}{6568}, -\frac{25\sqrt{4926}i}{6568}, -\frac{25\sqrt{4926}i}{6568}, \frac{115\sqrt{1642}i}{13136}, -\frac{115\sqrt{1642}i}{13136}, \frac{115\sqrt{1642}i}{13136}, -\frac{115\sqrt{1642}i}{13136}, -\frac{43\sqrt{4926}i}{13136}, \frac{43\sqrt{4926}i}{13136}, \frac{43\sqrt{4926}i}{13136}, -\frac{43\sqrt{4926}i}{13136}\right]$

y31 $\mathbb{Q}_{2,1}^{(s)}(T) = \left[-\frac{43\sqrt{4926}}{13136}, \frac{43\sqrt{4926}}{13136}, \frac{43\sqrt{4926}}{13136}, -\frac{43\sqrt{4926}}{13136}, \frac{25\sqrt{4926}}{6568}, \frac{25\sqrt{4926}}{6568}, -\frac{25\sqrt{4926}}{6568}, -\frac{25\sqrt{4926}}{6568}, \frac{115\sqrt{1642}}{13136}, -\frac{115\sqrt{1642}}{13136}, \frac{115\sqrt{1642}}{13136}, -\frac{115\sqrt{1642}}{13136}\right]$

y32 $\mathbb{Q}^{(s)}_{2,2}(T) = \left[\frac{115\sqrt{1642}}{13136}, -\frac{115\sqrt{1642}}{13136}, \frac{115\sqrt{1642}}{13136}, -\frac{115\sqrt{1642}}{13136}, -\frac{43\sqrt{4926}}{13136}, \frac{43\sqrt{4926}}{13136}, \frac{43\sqrt{4926}}{13136}, -\frac{43\sqrt{4926}}{13136}, \frac{25\sqrt{4926}}{6568}, \frac{25\sqrt{4926}}{6568}, -\frac{25\sqrt{4926}}{6568}, -\frac{25\sqrt{4926}}{6568}\right]$

y33 $\mathbb{Q}^{(s)}_{2,3}(T) = \left[\frac{25\sqrt{4926}}{6568}, \frac{25\sqrt{4926}}{6568}, -\frac{25\sqrt{4926}}{6568}, -\frac{25\sqrt{4926}}{6568}, \frac{115\sqrt{1642}}{13136}, -\frac{115\sqrt{1642}}{13136}, \frac{115\sqrt{1642}}{13136}, -\frac{115\sqrt{1642}}{13136}, -\frac{43\sqrt{4926}}{13136}, \frac{43\sqrt{4926}}{13136}, \frac{43\sqrt{4926}}{13136}, -\frac{43\sqrt{4926}}{13136}\right]$

y34 $\mathbb{Q}^{(s)}_{3,1}(T,1) = \left[\frac{5\sqrt{821}}{3284}, -\frac{5\sqrt{821}}{3284}, -\frac{5\sqrt{821}}{3284}, \frac{5\sqrt{821}}{3284}, -\frac{11\sqrt{821}}{821}, -\frac{11\sqrt{821}}{821}, \frac{11\sqrt{821}}{821}, \frac{11\sqrt{821}}{821}, \frac{21\sqrt{2463}}{3284}, -\frac{21\sqrt{2463}}{3284}, \frac{21\sqrt{2463}}{3284}, -\frac{21\sqrt{2463}}{3284}\right]$

y35 $\mathbb{Q}^{(s)}_{3,2}(T,1) = \left[\frac{21\sqrt{2463}}{3284}, -\frac{21\sqrt{2463}}{3284}, \frac{21\sqrt{2463}}{3284}, -\frac{21\sqrt{2463}}{3284}, \frac{5\sqrt{821}}{3284}, -\frac{5\sqrt{821}}{3284}, -\frac{5\sqrt{821}}{3284}, \frac{5\sqrt{821}}{3284}, -\frac{11\sqrt{821}}{821}, -\frac{11\sqrt{821}}{821}, \frac{11\sqrt{821}}{821}, \frac{11\sqrt{821}}{821}\right]$

y36 $\mathbb{Q}^{(s)}_{3,3}(T,1) = \left[-\frac{11\sqrt{821}}{821}, -\frac{11\sqrt{821}}{821}, \frac{11\sqrt{821}}{821}, \frac{11\sqrt{821}}{821}, \frac{21\sqrt{2463}}{3284}, -\frac{21\sqrt{2463}}{3284}, \frac{21\sqrt{2463}}{3284}, -\frac{21\sqrt{2463}}{3284}, \frac{5\sqrt{821}}{3284}, -\frac{5\sqrt{821}}{3284}, -\frac{5\sqrt{821}}{3284}, \frac{5\sqrt{821}}{3284}\right]$

Site and Bond

Table 5: Orbital of each site

#	site	orbital
1	Co	$\|s\rangle$, $\|p_x\rangle$, $\|p_y\rangle$, $\|p_z\rangle$, $\|d_v\rangle$, $\|d_{xy}\rangle$, $\|d_{xz}\rangle$, $\|d_{yz}\rangle$, $\|d_u\rangle$
2	Si	$\|s\rangle$, $\|p_x\rangle$, $\|p_y\rangle$, $\|p_z\rangle$

Table 6: Neighbor and bra-ket of each bond

#	head	tail	neighbor	head (bra)	tail (ket)
1	Co	Co	[1,2,3]	[s,p,d]	[s,p,d]

continued ...

Table 6

#	head	tail	neighbor	head (bra)	tail (ket)
2	Co	Si	[1,2,3]	[s,p,d]	[s,p]
3	Si	Si	[1,2,3]	[s,p]	[s,p]

Site in Unit Cell

Sites in (conventional) cell (no plus set), SL = sublattice

Table 7: 'Co' (#1) site cluster (4a), .3.

SL	position ($\boldsymbol{s}$)	mapping
1	[0.14356, 0.14356, 0.14356]	[1,5,9]
2	[0.35644, 0.85644, 0.64356]	[2,7,12]
3	[0.85644, 0.64356, 0.35644]	[3,8,10]
4	[0.64356, 0.35644, 0.85644]	[4,6,11]

Table 8: 'Si' (#2) site cluster (4a), .3.

SL	position ($\boldsymbol{s}$)	mapping
1	[0.84354, 0.84354, 0.84354]	[1,5,9]
2	[0.65646, 0.15646, 0.34354]	[2,7,12]

continued ...

Table 8

SL	position ($\boldsymbol{s}$)	mapping
3	[0.15646, 0.34354, 0.65646]	[3,8,10]
4	[0.34354, 0.65646, 0.15646]	[4,6,11]

Bond in Unit Cell

Bonds in (conventional) cell (no plus set): tail, head = (SL, plus set), (N)D = (non)directional (listed up to 3th neighbor at most)

Table 9: 1-th 'Co'-'Co' [1] (#1) bond cluster (12a@12b), D, $|\boldsymbol{v}|$= 2.72805 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[0.21287,-0.28713,-0.50000]	[0.25000, 0.00000, 0.89356]	[1]	(2,1)	(1,1)	[0,1,1]
2	[-0.21287, 0.28713,-0.50000]	[0.25000, 0.00000, 0.39356]	[2]	(1,1)	(2,1)	[0,-1,0]
3	[-0.21287,-0.28713, 0.50000]	[0.75000, 0.50000, 0.60644]	[3]	(4,1)	(3,1)	[0,0,0]
4	[0.21287, 0.28713, 0.50000]	[0.75000, 0.50000, 0.10644]	[4]	(3,1)	(4,1)	[0,0,-1]
5	[-0.50000, 0.21287,-0.28713]	[0.89356, 0.25000, 0.00000]	[5]	(4,1)	(1,1)	[1,0,1]
6	[-0.50000,-0.21287, 0.28713]	[0.39356, 0.25000, 0.00000]	[6]	(1,1)	(4,1)	[0,0,-1]
7	[0.50000,-0.21287,-0.28713]	[0.60644, 0.75000, 0.50000]	[7]	(3,1)	(2,1)	[0,0,0]
8	[0.50000, 0.21287, 0.28713]	[0.10644, 0.75000, 0.50000]	[8]	(2,1)	(3,1)	[-1,0,0]
9	[-0.28713,-0.50000, 0.21287]	[0.00000, 0.89356, 0.25000]	[9]	(3,1)	(1,1)	[1,1,0]
10	[0.28713,-0.50000,-0.21287]	[0.00000, 0.39356, 0.25000]	[10]	(1,1)	(3,1)	[-1,0,0]
11	[-0.28713, 0.50000,-0.21287]	[0.50000, 0.60644, 0.75000]	[11]	(2,1)	(4,1)	[0,0,0]

continued ...

Table 9

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
12	[0.28713, 0.50000, 0.21287]	[0.50000, 0.10644, 0.75000]	[12]	(4,1)	(2,1)	[0,-1,0]

Table 10: 2-th 'Co'-'Co' [1] (#2) bond cluster (12a@12b), D, $|\boldsymbol{v}|$= 3.97868 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[0.21287, 0.71287,-0.50000]	[0.25000, 0.50000, 0.89356]	[1]	(2,1)	(1,1)	[0,0,1]
2	[-0.21287,-0.71287,-0.50000]	[0.25000, 0.50000, 0.39356]	[2]	(1,1)	(2,1)	[0,0,0]
3	[-0.21287, 0.71287, 0.50000]	[0.75000, 0.00000, 0.60644]	[3]	(4,1)	(3,1)	[0,-1,0]
4	[0.21287,-0.71287, 0.50000]	[0.75000, 0.00000, 0.10644]	[4]	(3,1)	(4,1)	[0,1,-1]
5	[-0.50000, 0.21287, 0.71287]	[0.89356, 0.25000, 0.50000]	[5]	(4,1)	(1,1)	[1,0,0]
6	[-0.50000,-0.21287,-0.71287]	[0.39356, 0.25000, 0.50000]	[6]	(1,1)	(4,1)	[0,0,0]
7	[0.50000,-0.21287, 0.71287]	[0.60644, 0.75000, 0.00000]	[7]	(3,1)	(2,1)	[0,0,-1]
8	[0.50000, 0.21287,-0.71287]	[0.10644, 0.75000, 0.00000]	[8]	(2,1)	(3,1)	[-1,0,1]
9	[0.71287,-0.50000, 0.21287]	[0.50000, 0.89356, 0.25000]	[9]	(3,1)	(1,1)	[0,1,0]
10	[-0.71287,-0.50000,-0.21287]	[0.50000, 0.39356, 0.25000]	[10]	(1,1)	(3,1)	[0,0,0]
11	[0.71287, 0.50000,-0.21287]	[0.00000, 0.60644, 0.75000]	[11]	(2,1)	(4,1)	[-1,0,0]
12	[-0.71287, 0.50000, 0.21287]	[0.00000, 0.10644, 0.75000]	[12]	(4,1)	(2,1)	[1,-1,0]

Table 11: 3-th 'Co'-'Co' [1] (#3) bond cluster (12a@12b), D, $|\boldsymbol{v}|$= 4.33079 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[-0.78713,-0.28713,-0.50000]	[0.75000, 0.00000, 0.89356]	[1]	(2,1)	(1,1)	[1,1,1]
2	[0.78713, 0.28713,-0.50000]	[0.75000, 0.00000, 0.39356]	[2]	(1,1)	(2,1)	[-1,-1,0]
3	[0.78713,-0.28713, 0.50000]	[0.25000, 0.50000, 0.60644]	[3]	(4,1)	(3,1)	[-1,0,0]
4	[-0.78713, 0.28713, 0.50000]	[0.25000, 0.50000, 0.10644]	[4]	(3,1)	(4,1)	[1,0,-1]
5	[-0.50000,-0.78713,-0.28713]	[0.89356, 0.75000, 0.00000]	[5]	(4,1)	(1,1)	[1,1,1]
6	[-0.50000, 0.78713, 0.28713]	[0.39356, 0.75000, 0.00000]	[6]	(1,1)	(4,1)	[0,-1,-1]
7	[0.50000, 0.78713,-0.28713]	[0.60644, 0.25000, 0.50000]	[7]	(3,1)	(2,1)	[0,-1,0]
8	[0.50000,-0.78713, 0.28713]	[0.10644, 0.25000, 0.50000]	[8]	(2,1)	(3,1)	[-1,1,0]
9	[-0.28713,-0.50000,-0.78713]	[0.00000, 0.89356, 0.75000]	[9]	(3,1)	(1,1)	[1,1,1]
10	[0.28713,-0.50000, 0.78713]	[0.00000, 0.39356, 0.75000]	[10]	(1,1)	(3,1)	[-1,0,-1]
11	[-0.28713, 0.50000, 0.78713]	[0.50000, 0.60644, 0.25000]	[11]	(2,1)	(4,1)	[0,0,-1]
12	[0.28713, 0.50000,-0.78713]	[0.50000, 0.10644, 0.25000]	[12]	(4,1)	(2,1)	[0,-1,1]

Table 12: 1-th 'Co'-'Si' [1] (#4) bond cluster (4a@4a), D, $|\boldsymbol{v}|$= 2.30656 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[0.30003, 0.30003, 0.30003]	[0.99355, 0.99355, 0.99355]	[1,5,9]	(1,1)	(1,1)	[-1,-1,-1]
2	[-0.30003,-0.30003, 0.30003]	[0.50645, 0.00645, 0.49355]	[2,7,12]	(2,1)	(2,1)	[0,1,0]
3	[-0.30003, 0.30003,-0.30003]	[0.00645, 0.49355, 0.50645]	[3,8,10]	(3,1)	(3,1)	[1,0,0]

continued ...

Table 12

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
4	[0.30003,-0.30003,-0.30003]	[0.49355, 0.50645, 0.00645]	[4,6,11]	(4,1)	(4,1)	[0,0,1]

Table 13: 2-th 'Co'-'Si' [1] (#5) bond cluster (12a@12b), D, $|\boldsymbol{v}|$= 2.33786 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[-0.48710, 0.01290,-0.19997]	[0.59999, 0.84999, 0.74355]	[1]	(2,1)	(1,1)	[0,0,0]
2	[0.48710,-0.01290,-0.19997]	[0.90001, 0.15001, 0.24355]	[2]	(1,1)	(2,1)	[-1,0,0]
3	[0.48710, 0.01290, 0.19997]	[0.40001, 0.34999, 0.75645]	[3]	(4,1)	(3,1)	[0,0,0]
4	[-0.48710,-0.01290, 0.19997]	[0.09999, 0.65001, 0.25645]	[4]	(3,1)	(4,1)	[1,0,0]
5	[-0.19997,-0.48710, 0.01290]	[0.74355, 0.59999, 0.84999]	[5]	(4,1)	(1,1)	[0,0,0]
6	[-0.19997, 0.48710,-0.01290]	[0.24355, 0.90001, 0.15001]	[6]	(1,1)	(4,1)	[0,-1,0]
7	[0.19997, 0.48710, 0.01290]	[0.75645, 0.40001, 0.34999]	[7]	(3,1)	(2,1)	[0,0,0]
8	[0.19997,-0.48710,-0.01290]	[0.25645, 0.09999, 0.65001]	[8]	(2,1)	(3,1)	[0,1,0]
9	[0.01290,-0.19997,-0.48710]	[0.84999, 0.74355, 0.59999]	[9]	(3,1)	(1,1)	[0,0,0]
10	[-0.01290,-0.19997, 0.48710]	[0.15001, 0.24355, 0.90001]	[10]	(1,1)	(3,1)	[0,0,-1]
11	[0.01290, 0.19997, 0.48710]	[0.34999, 0.75645, 0.40001]	[11]	(2,1)	(4,1)	[0,0,0]
12	[-0.01290, 0.19997,-0.48710]	[0.65001, 0.25645, 0.09999]	[12]	(4,1)	(2,1)	[0,0,1]

Table 14: 3-th 'Co'-'Si' [1] (#6) bond cluster (12a@12b), D, $|\boldsymbol{v}|$= 2.44413 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[0.51290, 0.01290,-0.19997]	[0.09999, 0.84999, 0.74355]	[1]	(2,1)	(1,1)	[-1,0,0]
2	[-0.51290,-0.01290,-0.19997]	[0.40001, 0.15001, 0.24355]	[2]	(1,1)	(2,1)	[0,0,0]
3	[-0.51290, 0.01290, 0.19997]	[0.90001, 0.34999, 0.75645]	[3]	(4,1)	(3,1)	[1,0,0]
4	[0.51290,-0.01290, 0.19997]	[0.59999, 0.65001, 0.25645]	[4]	(3,1)	(4,1)	[0,0,0]
5	[-0.19997, 0.51290, 0.01290]	[0.74355, 0.09999, 0.84999]	[5]	(4,1)	(1,1)	[0,-1,0]
6	[-0.19997,-0.51290,-0.01290]	[0.24355, 0.40001, 0.15001]	[6]	(1,1)	(4,1)	[0,0,0]
7	[0.19997,-0.51290, 0.01290]	[0.75645, 0.90001, 0.34999]	[7]	(3,1)	(2,1)	[0,1,0]
8	[0.19997, 0.51290,-0.01290]	[0.25645, 0.59999, 0.65001]	[8]	(2,1)	(3,1)	[0,0,0]
9	[0.01290,-0.19997, 0.51290]	[0.84999, 0.74355, 0.09999]	[9]	(3,1)	(1,1)	[0,0,-1]
10	[-0.01290,-0.19997,-0.51290]	[0.15001, 0.24355, 0.40001]	[10]	(1,1)	(3,1)	[0,0,0]
11	[0.01290, 0.19997,-0.51290]	[0.34999, 0.75645, 0.90001]	[11]	(2,1)	(4,1)	[0,0,1]
12	[-0.01290, 0.19997, 0.51290]	[0.65001, 0.25645, 0.59999]	[12]	(4,1)	(2,1)	[0,0,0]

Table 15: 1-th 'Si'-'Si' [1] (#7) bond cluster (12a@12b), D, $|\boldsymbol{v}|$= 2.74663 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[-0.18708, 0.31292,-0.50000]	[0.75000, 0.00000, 0.59354]	[1]	(2,1)	(1,1)	[0,-1,0]
2	[0.18708,-0.31292,-0.50000]	[0.75000, 0.00000, 0.09354]	[2]	(1,1)	(2,1)	[0,1,1]
3	[0.18708, 0.31292, 0.50000]	[0.25000, 0.50000, 0.90646]	[3]	(4,1)	(3,1)	[0,0,-1]

continued ...

Table 15

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
4	[-0.18708,-0.31292, 0.50000]	[0.25000, 0.50000, 0.40646]	[4]	(3,1)	(4,1)	[0,0,0]
5	[-0.50000,-0.18708, 0.31292]	[0.59354, 0.75000, 0.00000]	[5]	(4,1)	(1,1)	[0,0,-1]
6	[-0.50000, 0.18708,-0.31292]	[0.09354, 0.75000, 0.00000]	[6]	(1,1)	(4,1)	[1,0,1]
7	[0.50000, 0.18708, 0.31292]	[0.90646, 0.25000, 0.50000]	[7]	(3,1)	(2,1)	[-1,0,0]
8	[0.50000,-0.18708,-0.31292]	[0.40646, 0.25000, 0.50000]	[8]	(2,1)	(3,1)	[0,0,0]
9	[0.31292,-0.50000,-0.18708]	[0.00000, 0.59354, 0.75000]	[9]	(3,1)	(1,1)	[-1,0,0]
10	[-0.31292,-0.50000, 0.18708]	[0.00000, 0.09354, 0.75000]	[10]	(1,1)	(3,1)	[1,1,0]
11	[0.31292, 0.50000, 0.18708]	[0.50000, 0.90646, 0.25000]	[11]	(2,1)	(4,1)	[0,-1,0]
12	[-0.31292, 0.50000,-0.18708]	[0.50000, 0.40646, 0.25000]	[12]	(4,1)	(2,1)	[0,0,0]

Table 16: 2-th 'Si'-'Si' [1] (#8) bond cluster (12a@12b), D, $|\boldsymbol{v}|$= 3.86202 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[-0.18708,-0.68708,-0.50000]	[0.75000, 0.50000, 0.59354]	[1]	(2,1)	(1,1)	[0,0,0]
2	[0.18708, 0.68708,-0.50000]	[0.75000, 0.50000, 0.09354]	[2]	(1,1)	(2,1)	[0,0,1]
3	[0.18708,-0.68708, 0.50000]	[0.25000, 0.00000, 0.90646]	[3]	(4,1)	(3,1)	[0,1,-1]
4	[-0.18708, 0.68708, 0.50000]	[0.25000, 0.00000, 0.40646]	[4]	(3,1)	(4,1)	[0,-1,0]
5	[-0.50000,-0.18708,-0.68708]	[0.59354, 0.75000, 0.50000]	[5]	(4,1)	(1,1)	[0,0,0]
6	[-0.50000, 0.18708, 0.68708]	[0.09354, 0.75000, 0.50000]	[6]	(1,1)	(4,1)	[1,0,0]

continued ...

Table 16

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
7	[0.50000, 0.18708,-0.68708]	[0.90646, 0.25000, 0.00000]	[7]	(3,1)	(2,1)	[-1,0,1]
8	[0.50000,-0.18708, 0.68708]	[0.40646, 0.25000, 0.00000]	[8]	(2,1)	(3,1)	[0,0,-1]
9	[-0.68708,-0.50000,-0.18708]	[0.50000, 0.59354, 0.75000]	[9]	(3,1)	(1,1)	[0,0,0]
10	[0.68708,-0.50000, 0.18708]	[0.50000, 0.09354, 0.75000]	[10]	(1,1)	(3,1)	[0,1,0]
11	[-0.68708, 0.50000, 0.18708]	[0.00000, 0.90646, 0.25000]	[11]	(2,1)	(4,1)	[1,-1,0]
12	[0.68708, 0.50000,-0.18708]	[0.00000, 0.40646, 0.25000]	[12]	(4,1)	(2,1)	[-1,0,0]

Table 17: 3-th 'Si'-'Si' [1] (#9) bond cluster (12a@12b), D, $|\boldsymbol{v}|$= 4.4386 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[-1.00000, 0.00000, 0.00000]	[0.34354, 0.84354, 0.84354]	[1]	(1,1)	(1,1)	[1,0,0]
2	[1.00000, 0.00000, 0.00000]	[0.15646, 0.15646, 0.34354]	[2]	(2,1)	(2,1)	[-1,0,0]
3	[1.00000, 0.00000, 0.00000]	[0.65646, 0.34354, 0.65646]	[3]	(3,1)	(3,1)	[-1,0,0]
4	[-1.00000, 0.00000, 0.00000]	[0.84354, 0.65646, 0.15646]	[4]	(4,1)	(4,1)	[1,0,0]
5	[0.00000,-1.00000, 0.00000]	[0.84354, 0.34354, 0.84354]	[5]	(1,1)	(1,1)	[0,1,0]
6	[0.00000, 1.00000, 0.00000]	[0.34354, 0.15646, 0.15646]	[6]	(4,1)	(4,1)	[0,-1,0]
7	[0.00000, 1.00000, 0.00000]	[0.65646, 0.65646, 0.34354]	[7]	(2,1)	(2,1)	[0,-1,0]
8	[0.00000,-1.00000, 0.00000]	[0.15646, 0.84354, 0.65646]	[8]	(3,1)	(3,1)	[0,1,0]
9	[0.00000, 0.00000,-1.00000]	[0.84354, 0.84354, 0.34354]	[9]	(1,1)	(1,1)	[0,0,1]

continued ...

Table 17

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
10	[0.00000, 0.00000, 1.00000]	[0.15646, 0.34354, 0.15646]	[10]	(3,1)	(3,1)	[0,0,-1]
11	[0.00000, 0.00000, 1.00000]	[0.34354, 0.65646, 0.65646]	[11]	(4,1)	(4,1)	[0,0,-1]
12	[0.00000, 0.00000,-1.00000]	[0.65646, 0.15646, 0.84354]	[12]	(2,1)	(2,1)	[0,0,1]